

VACUUM TUBE (THERMIONIC VALVE): An electron tube that is either evacuated or filled with low-pressure gas and in which electrons are pulled from the cathode by an applied anode voltage. It was used in early electronic circuitry to control a flow of electrons. In the first half of the 20th century, vacuum tubes allowed the development of radio broadcasting, long-distance telephone service and television. It has been replaced by transistors in most applications and are still occasionally used in display devices for television sets and computers (cathode-ray tubes), in microwave ovens and as high-frequency transmitters on space satellites. The Diode Valve is so called because it has just two electrodes – the cathode and the anode. The Cathode is the electrode that is heated and emit electrons and the anode is the electrode that collects the electrons. Since electrons are negatively charged particles, they will only be attracted to the anode, if this is given a positive potential with respect to the cathode. This explains why the valve will only conduct in one direction, from cathode to anode and not vice versa. It also explains the choice of the word valve to describe the device since a valve, is by definition, a one-way device. The Americans call it a vacuum tube, due to this.

VAGAL TONE: The impulses from the vagus nerve producing inhibition of the heartbeat. This vagal tone reduces the resting heart rate down to 60-80 beats/min.

VAGINA: The passage leading from the uterus to the outside in females, which together with the cavity of the uterus, form the birth canal. Semen is deposited in the vagina during copulation. The vagina's lining thickens and thins during the menstrual cycle in response to oestrogen from the ovaries, being thickest and most elastic during ovulation and pregnancy. It is slightly acidic, thus discouraging infection by pathogens. It has thick, elastic muscle walls to accommodate movement of the penis during intercourse and passage of a child during delivery and through which mucus-like fluid seeps for lubrication during sexual arousal. In most virgins, its external opening is partially closed by a thin fold of tissue called **hymen**, which has various forms, not all of which significantly obstruct the opening.

VAGRANT: An individual organism found outside its known range of habitat. Vagrancy occurs in animals such as birds, insects, mammals and sea turtles.

VAGUS NERVE (PNEUMOGASTRIC NERVE): A paired nerve of the cranial nerves, passing through the neck and thorax into the abdomen, branching to many major internal organs. It supplies sensation to part of the ear, the tongue, the larynx and the pharynx; gives motor impulses to the vocal cords; and supplies motor and secretory impulses to the abdominal and thoracic viscera. It is the tenth and longest of the cranial nerves.

VALENCE (VALENCY): 1. Positive number that characterises the combining power of an atom of a given element to the number of hydrogen atoms or their equivalent in a chemical reaction. The valence also shows the combining power of atoms or groups and is measured by the number of electrons the atom or group will give up, receive or share, when forming a compound. For an ion, the valence equals the ionic charge in ionic compounds. For example, in Na_2S , sodium has a valency of 1 (Na^+) and sulfur a valency of 2 (S^{2-}). In covalent compounds, it is equal to the number of bonds formed. For example, in CO_2 , oxygen has a valency of 2 and carbon has a valency of 4. 2. The number of different antigenic determinants with which a single antibody molecule can combine.

VALENCE BAND: 1. The highest energy level in an insulator or semiconductor that can be filled with electrons. It is an electron band, which is located beneath the conduction band and contains the valence electrons of atoms. In a semiconductor, the electrons are too tightly bound to the atom to carry electric current. However, with a rise in temperature some of the electrons in the valence band gain thermal energy and cross the energy gap and move into the adjacent conduction band to conduct electricity. 2. Region of electronic energy level that binds atoms of a crystal together.

VALENCE BOND: The chemical bond formed by the interaction of valence electrons between two or more atoms.

VALENCE BOND THEORY (VBT): A theory of the structure of chemical compounds which postulates that a chemical bond is formed as a result of the overlap of the atomic orbitals of the participating atoms. Electrons used in the bonding are thus considered localised between the atoms involved in the bond region. The overlapping atomic orbitals according to this theory may be different and may not necessarily be 'pure' atomic orbitals. **Sigma bonds**, according to the VBT, results from head-on overlap of atomic orbitals, while **pi bond** is formed by the lateral (side-to-side) overlap of atomic orbitals.

VALENCE ELECTRON: An electron in one of the outer shells of an atom that takes part in forming chemical bonds and determines the chemical properties of an element. In a sodium atom, the number of valence electron is 1.

VALENCE SHELL: The outer electron shell for a given atom, containing one or more valence electrons that are available to form bonds with other atoms to create molecules. The number of electrons in this shell determines the conductivity of the atom. An atom is at its most stable (and therefore unreactive) when all its orbitals are full. On the other hand, an atom is least stable (and therefore most reactive) when its valence electron orbitals are not full. Therefore, an element that has a full valence energy level is more stable and less likely to react than elements with a valence energy level that is not full.

VALENCE TAUTOMERISATION/TAUTOMERIZATION: A simple reversible and generally rapid isomerisations or degenerate rearrangements involving the formation and rupture of single and/or double bonds, without migration of atoms or groups.

VALENCE TRANSITION: A transition observed in certain rare-earth and actinide materials in which the electronic occupation of the 4f or 5f orbital changes with external conditions for example, temperature and pressure. An example is the transition at approximately 0.65 GPa, where black, semiconducting $\text{Sm}^{2+}\text{S}^{2-}$ changes to golden metallic $(\text{Sm}^{3+}+\text{e}^-)\text{S}^{2-}$.

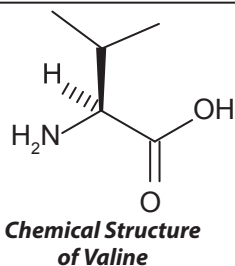
VALENCE-SHELL ELECTRON PAIR REPULSION MODEL (VSEPR; MODEL VALENCE-SHELL ELECTRON PAIR REPULSION THEORY): A model that accounts for the geometrical arrangements of shared and unshared electron pairs around a central atom in terms of the repulsions between electron pairs.

VALENCE-SHELL EXPANSION (EXPANDED OCTET; EXPANDED VALENCE SHELL): The use of d orbitals in addition to s and p orbitals to form a covalent bond; or atoms that form molecules even though they have more than 8 valence electrons. For example, sulfur can promote some of its electrons into d orbitals and increase its valence. Any element that contains d electrons can theoretically expand its valence shell. Atoms such as the noble gases except helium that already have 8 valence electrons do not react with other atoms, because they are already in their most stable configuration. Atoms that do not already have 8 valence electrons either transfer or share electrons to become stable.

VALID: 1. In logic, it is having the conclusion following logically from the premise so that whenever the premise is true the conclusion must also be true. 2. Of a sentence in a formal language, true in every interpretation; satisfied by every assignment of values to the variables in every interpretation, so that every interpretation is a model for the statement.

VALIDATION: In computer science, it is the process of checking input data to ensure that it is complete, accurate and falls within the accepted boundaries of the application collecting the data. Validation procedures are typically written into the program code and ensure that most errors are detected.

VALINE (2-AMINO-3-METHYLBUTANOIC ACID): One of the essential amino acids with the chemical formula $C_5H_{11}NO_2$, density 1.316 g/cm^3 , melting point of 298°C (571 K) and decomposes. It is found in most proteins required for normal growth in animals. Human dietary sources include cottage cheese, fish, poultry, peanuts, sesame seeds and lentils. It is produced industrially by hydrolysis of proteins or by chemical synthesis. It is used in biochemical and nutritional research and as a dietary supplement.



VALLECULAR CANAL: A longitudinal channel in the stem internode of *Equisetum* and some of its fossil relatives, positioned radially opposite a longitudinal furrow (valley) between the stem ridges. The vallecular canals are arranged roughly between the vascular bundles.

VALLEY GLACIER (ALPINE GLACIER): A glacier located on top of a mountain that moves downhill through the valley.

VALLEY OF STABILITY: The bottom region of the surface defined by plotting the numbers of protons and neutrons of a nucleus horizontally and the energy per nucleon of the nucleus vertically. Only about 300 nuclear species are stable out of thousands, existing along the so-called "valley of stability". The unstable species forming the valley "walls" are those with an overabundance of protons or neutrons and tend to decay quickly, sometimes within milliseconds. The concept of valley of stability enables several important topics in nuclear physics such as nuclear fission and nuclear fusion to be understood in a clear physical way.

VALLEY TRAIN: An accumulation of fluvioglacial deposits laid down between valley walls by meltwaters far beyond the margin of an active glacier. It is a type of outwash.

VALUATION: In logic, a function that assigns truth values (true or false) to individual propositional variables in any logical expression. A **valuation system** is a set of values that are assigned to sentences by a valuation, together with a set of designated values. For example, in bivalent logic, the valuation system is $\{IT, F\}, \{T, I\}$.

VALUE: 1. An assigned or calculated numerical quantity. 2. The magnitude of a particular quantity generally expressed as a unit of measurement multiplied by a number. 3. The particular quantity that is the result of applying a function or operation, for some given argument. An example is the value of the function $y = x^2$ for $x = 3$ is 9. Also, it is an assignment of a significance to a variable; for example, for any value of x , $x = x$; the value of x for which $3x = 6$ is 2. 4. The value assigned to a statement by a valuation; a sentential variable in classical logic may take either of the values true and false.

VALUE OF A DIVISION: The reciprocal of the sensitivity of a precision balance scale, which usually varies with load. It is determined by empirical calibration.

VALUE-ADDED NETWORK (VAN): A simple communications network that offers additional services such as message routing and stock quote services.

VALUE-ADDED TRAIT: A quality trait or characteristic that increases the value of a crop product relative to its typical or commodity version, and that requires a uniform and uncontaminated end product. In soybean, it is commonly referred to as specialty varieties with specific physical or chemical characteristics that are required for specific markets.

VALVATE: 1. Describing sepals and petals that meet but do not overlap in the bud. 2. Opening by valves, as in many dehiscent fruits.

VALVE: 1. A membranous structure in a hollow organ or passage, as in an artery or vein, that folds or closes to prevent the return flow of the body fluid passing through it. Valves in the heart, veins and lymphatic vessels consist of two or three flaps of tissue fastened to

the walls. 2. Any of the parts that make up a capsule or other dry fruit that sheds its seeds. 3. One of the two halves of the cell wall of a diatom. 4. Either of the two hinged portions of the shell of a bivalve mollusc.

VALVULAR: In botany, it refers to dehiscing through flap-covered pores.

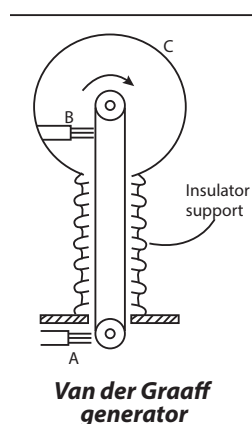
VALVULE (PALE; PALEA): The upper of the pair of bracts beneath each floret (flower) in the inflorescence of a grass, the other being the lemma. It is usually thin, narrow and parallel sided. It usually has two ribs, which may project as prominent keels.

VAMPIRE BAT: South American bat of the genus *Desmodus* that feeds on blood of other animals. Such behaviour is called haematophagy.

VAMPIRE NUMBER: A natural number x that can be factorised as $y \times z$ in such a way that the number of occurrences of a particular digit in the representation of x in a given base, for example, 10 appears the same number of times in the representations in that same base of y and z together. For example, 2187 is a vampire number since $2187 = 21 \times 87$; similarly 136948 is a vampire because $136948 = 146 \times 938$.

VAN ALLEN BELTS (RADIATION BELTS): Either of the two zones of high-intensity particulate radiation trapped in the Earth's magnetic field and surrounding the planet. The lower belt extends from 1000 to 5000 km above the equator, contains electrons and protons, while the upper belt extends from 15000 to 25000 km above the equator, containing mainly electrons. They were discovered by American physicist, James Van Allen.

VAN DER GRAFF GENERATOR: An electrostatic generator that can produce high voltage, usually in the range of megavolts in which an electric charge is either removed from or transferred to a large hollow spherical electrode by a rapidly moving belt, accelerating particles to energies of about ten mega electron volts.. It consists essentially of an endless insulated fabric belt moving vertically. A charge is applied to the belt by the needlepoints A from an external source. This charge is carried continuously up into a large hollow sphere C, where collector point B removes it.



VAN DER POL EQUATION: The differential equation $u''(u^2 - 1)u' + \beta u = 0$, which has exactly one periodic solution.

VAN DER WAAL'S EQUATION: A modified equation of state proposed by J. D. van der Waals in 1873 describing the physical behaviour of gases that takes into account the volumes of molecules and the interactions between them. It explains the difference in behaviour between a real gas and an ideal gas that obeys the gas laws. The equation is $(p + \frac{n^2 a}{v^2})(v - nb) = nRT$, where P is the pressure gas, v the total volume of the container containing the gas, a is a measure of the attraction between the particles, n is the number of moles, b the volume excluded by a mole of particles, R is gas constant and T is absolute temperature.

VAN DER WAAL'S FORCES (VAN DER WAAL'S ATTRACTION): A weak attractive force between atoms or molecules named after Johannes van der Waals (1837–1923), resulting from the positioning of the electrons within the interacting particles or weak forces between atoms, caused by the movement of electrons within the atoms. The force accounts for the term a/V^2 in van der Waals' equation. These forces are much weaker than those arising from valence bonds and are inversely proportional to the seventh power of the distance

between the atoms or molecules. Electrodynamics forces arise between atoms, molecules and assemblies of molecules due to their vibrations, giving rise to electromagnetic interactions. These are attractive when the vibrational frequencies and absorptions are identical or similar, repulsive when they are not identical. There are three factors causing such forces: (i) dipole-dipole interaction, i.e. electrostatic attractions between two molecules with permanent dipole moments; (ii) dipole-induced dipole interactions, in which the dipole of one molecule polarises a neighbouring molecule; and (iii) dispersion forces arising because of small instantaneous dipoles in atoms.

VAN DER WAERDEN'S CONJECTURE: The conjectured that the unique Doubly stochastic $n \times n$ matrix of minimum permanent (whose value is $n! \times n^{-n}$) is the constant matrix with all entries $1/n$.

VAN NIEL, CORNELIUS BERNARDUS (1897-1985): Dutch-American microbiologist, who introduced the study of general microbiology to the United States and made key discoveries explaining the chemistry of photosynthesis.

VAN'T HOFF FACTOR: A factor appearing in equations for colligative properties predicted by Van't Hoff's law. It is the ratio of the number of actual particles present to the number of undissociated particles. Its symbol is i .

VAN'T HOFF ISOCHORE: A thermodynamic equation for the variation with temperature T of the equilibrium constant K of a gaseous reaction in terms of the heat of reaction at constant pressure, derived by Jacobus Van't Hoff. It is given as $(d \log K) / dT = \Delta H / RT^2$, where T is the absolute temperature, ΔH the enthalpy of the reaction, K is the equilibrium constant and R is the gas constant.

VANADIUM: A silvery-white malleable, ductile and poisonous metallic transition element found in several minerals, notably vanadinite and carnotite with the symbol V, atomic number 23, atomic weight 50.942, melting point 1890 °C (2163 K), boiling point 3000 °C (3273 K) and relative density 6.11. It is used in the production of rust-resistant high-speed tools, as a carbon stabiliser in some steels, as a titanium-steel bonding agent and as a catalyst.

VANADIUM(V) OXIDE (VANADIUM PENTOXIDE; DIVANADIUM; VANADIC ANHYDRIDE; DIVANADIUM PENTOXIDE): A poisonous yellow or red crystalline compound with the chemical formula V_2O_5 , density 3.357 g/cm³, melting point 690 °C (963 K), boiling point 1750 °C (2023 K) and decomposes. It is produced when vanadium metal is heated with excess oxygen. In the laboratory, preparation involves the decomposition of ammonium metavanadate at around 200 °C (473 K): $2NH_4VO_3 \rightarrow V_2O_5 + 2NH_3 + H_2O$. It is used in the manufacture of glass and as a catalyst in various organic reactions and as a starting material for other vanadium salts.

VANDERMONDE

DETERMINANT: In mathematics, it is the determinant of the square matrix of which each row consists of the zeroth to $(n-1)$ th powers of one of n numbers as shown in the matrix.

$$\begin{vmatrix} 1 & x_1 & \dots & x_1^{n-1} \\ 1 & x_2 & \dots & x_2^{n-1} \\ \vdots & \vdots & \dots & \vdots \\ 1 & x_n & \dots & x_n^{n-1} \end{vmatrix} = \prod_{i < j} (x_i - x_j).$$

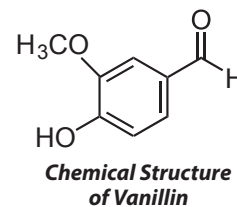
VANE: 1. A weather vane, which is a narrow pointer on top of a building turned by the wind so as to show the direction of the wind. 2. Any of several usually relatively thin, rigid, flat or sometimes curved surfaces radially mounted along an axis, as a blade in a turbine or a sail on a windmill, that is turned by or used to turn a fluid. 3. The flattened, web-like part of a feather, consisting of a series of barbs on either side of the shaft.

VANILLA: The long narrow fleshy seedpod of a tropical climbing orchid of the genus *Vanilla* in the orchid family, especially *V. Planifolia*. It is cultivated for its long narrow seedpods from which a flavouring agent is obtained. It can also be produced artificially. It is used as

flavouring in food or as perfumes. The compound vanillin, 4-hydroxy-3-methoxybenzaldehyde is primarily responsible for the characteristic flavour and smell of vanilla.

VANILLIN (4-HYDROXY-3-METHOXYBENZALDEHYDE; METHYL VANILLIN; VANILLIC ALDEHYDE): A white or yellowish

crystalline aldehyde with the chemical formula $C_8H_8O_3$, density 1.056 g cm⁻³, melting point 81-83 °C (354-356 K) and boiling point 285 °C (558 K). It has a taste and smell similar to vanilla. Vanillin is found in vanilla beans and certain balsams and resins. Synthetic vanillin is made either from guaiacol or from lignin, a constituent of wood, which is a by-product of the pulp industry. It is used in perfumes, flavourings and pharmaceuticals.



VANISH (VANISH IDENTICALLY): In mathematics, to take on the value of zero or tend to zero. For example, the function $f(z) = z^2$ vanishes mathematically at the point $z = 0$.

VANISH AT INFINITY: Of a continuous complex-valued function on a locally compact space, such that for every $a > 0$ there is a compact set K_a such that $|f(x)| < a$ for all x not in K_a .

VANISH NOWHERE: In mathematics, it refers to an algebra that satisfies the condition that given any point of the set, there is a member of the algebra for which the value at the point is non-zero; this happens if 1 is in the algebra.

VANISHING LINE: One of the parallel lines converging to the vanishing point in a picture or painting, it connects a point on a figure with the vanishing point. In a picture or painting, **vanishing point** is the point where parallel lines seem to meet at a distance.

VAPOUR: A substance in the gaseous state, which may be liquefied by increasing the pressure without altering the temperature (i.e., an isothermal increase in pressure).

VAPOUR DENSITY: The density of a gas or vapour relative to hydrogen, oxygen or air. In the case of the vapour density it is the ratio of the mass of a particular volume of a gas to the mass of an equal volume of hydrogen under identical conditions of pressure and temperature. Taking the density of hydrogen as 1, this ratio is equal to half the relative molecular mass of the gas. When defined with respect to air, the air is given an arbitrary vapour density of 1. With this definition, the vapour density would indicate whether a gas is denser (greater than 1) or less dense (less than 1) than air. This has important implication in container storage and personal safety. Dense gas could sink and so collect on the ground and if flammable, could be hazardous.

VAPOUR DRIFT: The movement of pesticide or herbicidal vapours from the area of application due to chemical volatilisation of the product. Several factors are important in determining how much spray will drift and these primarily involve weather conditions and sprayer setup. Spray breaks up into droplets of varying size with larger droplets falling onto the target area, while the smallest droplets may remain in the air as vapours. Because droplet drift usually disperses as it moves away from the sprayed area, it causes crop damage in adjoining areas. The damage decreases as the distance from the point of application increases.

VAPOUR PHASE INTERFERENCE: Interference caused by a change in the fraction of analyte dissociated, ionised or excited in the gaseous phase.

VAPOUR PRESSURE: The saturation pressures exerted by the vapour given off by a liquid or solid, particularly a vapour in contact with its liquid form. It is one of the most important physical properties of a liquid and the vapour pressure, used in many thermodynamic calculations and underlies several methods for the determination of the molecular weights of substances dissolved in liquids. It is caused by vibrating atoms or molecules continuously escaping from its surface.

VAPOURISATION TEMPERATURE (VAPORIZATION TEMPERATURE): The temperature of the atomisation surface at which analyte loss becomes statistically significant.

VAPOURWARE (VAPORWARE): Software that has been officially announced by a company, but has not officially been released several months or years after the official announced release date.

VARIABLE: 1. Generally, it defines anything that is likely to change, vary or changeable; having no fixed quantitative value. 2. In mathematics, a variable is a symbol or letter, such as "x" or "y," that represents a value. In algebraic equations, the value of one variable is often dependent on the value of another. For example, in the equation $y = 5 + 2x$, y is the **dependent variable** because its value is based on the value assigned to the independent variable x . The equation may also be called a **function** since y is a function of x . If $x = 1$, then $y = 12$. If $x = 2$, then, then $y = 14$. 3. In statistics, it is the characteristic measured or observed when an experiment is carried out or an observation is made. Variables may be numerical or non-numerical and since a non-numerical observation may be coded numerically, a variable is taken usually to be numerical, for statistics, is always concerned with random variables and variables whose measurement may involve random errors. 4. In computer science, it is a symbolic storage location uniquely named by the program that holds the same data assessed to it until a new value is given. It references a memory location.

VARIABLE ANTIGEN (ANTIGENIC VARIATION): The process by which surface antigens of parasitic or pathogenic organisms can alter their antigenic character to evade a host's immune responses. Antigenic variation also allows pathogens to cause re-infection as they are not recognised by the host's immune system. When an organism is exposed to a particular antigen an immune response is stimulated and antibodies are generated to target that specific antigen.

VARIABLE NUMBER TANDEM REPEATS, VNTR (MINISATELLITE DNA): A family of genetic loci found in eukaryotes consisting of short sequence of DNA repeated in tandem arrays. In humans these arrays are typically 1-5 kb long. VNTR can be found on many chromosomes and often show variations in length between individuals; each variant acts as an inherited allele, allowing them to be used for personal or parental identification and DNA fingerprinting.

VARIABLE PATHLENGTH CELL: A cell whose path length can be varied either continuously or in steps by means of spacers.

VARIABLE RATE APPLICATION: In agriculture, it is the application of nutrients at differing rates on specific areas of an individual field based on the specific and differing requirements of those areas.

VARIABLE STAR: A star that varies in brightness. The brightness variation can be regular or irregular. Changes in brightness of a star occur both in its early stages and close to its death. Explosive variables include novae and supernovae, which brighten rapidly due to sudden outbursts of radiant energy. The increased brightness lasts a short time, followed by relatively slow dimming. Eclipsing variable stars are variable only because light from one star is blocked by another in Earth's direction. Observations of variable stars provide information concerning the mass, size, temperature, structure and evolution of stars. There are more than 25,000 variable stars known, some of which can be seen with the naked eye. The time-scale from maximum to minimum brightness varies from less than a second to a few years.

VARIABLE TERM: A term that contains at least one variable. For example, in $5x - 8 = 2x + 5$, $5x$ and $2x$ are variable terms.

VARIANCE: 1. A statistical measure of the standard deviation or the spread of data or variation of a group of numbers in a sample. It is equal to the square of the standard deviation or the ratio of the squared standard deviation to the sample size population variance or the ratio of the squared standard deviation to the sample size minus one sample variance. Population and sample variance are denoted

by σ^2 or s^2 respectively. Variance is given by $\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$, where

x_i is the various observations, μ is the mean and σ^2 is the variance.

Mean is given by $\mu = \frac{\sum_{i=1}^N x_i}{N}$. Unlike standard deviation, variances are

additive. Variance is never negative but it can be zero, only if the data values are the same. A small variance indicates that the data points tend to be very close to the mean (expected value) and hence to each other, while a high variance indicates that the data points are very spread out around the mean and from each other. 2. The number of thermodynamic variables, such as temperature and pressure, required to specify a state of equilibrium of a system, given by the phase rule.

VARIANCE-COVARIANCE MATRIX (COVARIANCE MATRIX):

A square matrix, usually denoted Σ in which variance $[\text{Var}(X_i)]$ and covariance $\text{Cov}(X_i, X_j)$ are displayed with the variances appearing along the main diagonal and the off-diagonal elements (those in the i, j position) contain the covariances between all possible pairs of variables. The matrix is then symmetrical and non-negative definite. If $X_1, X_2, \dots, X_n$ are random variables then the variance-covariance matrix is given by $\Sigma_{ij} = \text{cov}(x_i, x_j) = [(x_i - \mu_i)(x_j - \mu_j)]$.

VARIATE: In statistics, it is a random variable or a numerical value taken by it that obeys a given probabilistic law.

VARIATION: 1. The degree to which something differs, such as from a former state or value, from others of the same type or from a standard. 2. The differences in characteristics between individuals of the same species or a significant deviation from the normal biological form, function or structure or a living organism that differs from the normal form for its kind. Variations may be due to environmental conditions such as water supply or light intensity affecting the height of a leaf size of a plant; or genetic variation which may be due to differences in genetic constitution, recombination and increased by outbreeding. Genetic variation is transmittable while those caused by the environment are not. Wide genetic variation improves survival potential of a species. There are two main types: (i) **continuous variation** describes a characteristic that can be expressed across a whole range of values, for example, height and weight. There are intermediate forms between one extreme end of the character and the next and a gradual change in the character as one moves from one organism to the next. For example, in human beings, height, weight, hand span and foot size exist in continuous variation. (ii) **Discontinuous variation** describes characteristics which are either present or absent in a clear-cut form, for example, blood groups in humans. An individual can have only one type of blood group, there is no intermediate blood group. 3. A mathematical function that relates the values of one variable to those of other variables. 4. The supremum of the oscillations of a function over all finite partitions of a given interval. 5. The angle between magnetic north and true north at a particular point on the Earth's surface. 6. A change in or deviation from the average motion or orbit of an astronomical object; or a term representing the gravitational attraction of the Sun on the Earth-Moon system in the mathematical equation for the Moon's motion.

VARIATION OF AN INTEGRAL: The directional derivative of the integral $I(x) = \int_a^b f(x, x', t) dt$ in the direction of an admissible variation

h ; explicitly, $\delta I(x, h) = \lim_{\lambda \rightarrow 0} \frac{I(x + \lambda h) - I(x)}{\lambda}$, second- and higher-order variations are defined by taking higher derivatives of $I(x + \lambda h)$.

VARIATION OF PARAMETERS: 1. The method of finding a solution to an inhomogeneous linear differential equation, $L(x) = f$ by first finding a fundamental set of solutions $\{x_1, \dots, x_n\}$ for the homogeneous equation $L(x) = 0$ and then attempting to solve the equation

$L\left(\sum_{i=1}^n c_i x_i\right) = f$ for the undetermined functions c_i . With the condition that for $0 \leq k < n-1$, the derivative vanish, $\sum_{i=1}^n c_i x_i^{(k)}$ and differential

equation becomes $\sum_{i=1}^n c_i x_i^{(n-1)} = f$, for which n number of equations in

m, n unknowns is obtained. A unique solution for $\frac{dc_i(t)}{dt}$ and integration

will provide the desired solution. 2. For inhomogeneous systems of linear differential equation, $y' = A(t)y + b(t)$, the solution $y = \Omega(t)c + \int^t \Omega^{-1}(s)b(s)ds$, where $\Omega(t)$ is the principal solution matrix of $y' = A(t)y$.

VARIATIONAL CALCULUS (CALCULUS OF VARIATIONS): The branch of calculus concerned with finding the maxima or minima of definite integrals that depends for its values on another function to find functions that maximise or minimise a given function of those functions.

VARIATIONAL INEQUALITY: An inequality system in Hilbert space: given a nonlinear operator, T and a closed convex set, C , one seeks an x in C such that $\langle T(x), y - x \rangle \leq 0$ for all y in C . If the set is the entire space one finds a solution to $T(x) = 0$. If the set is the positive orthant in Euclidean space, this is a complementarity problem.

VARIATIONAL PRINCIPLE: A scientific principle used within the calculus of variations that if a trial wave function is used to calculate the energy of a system then the energy calculated cannot be lower than the actual energy of the ground state of the system.

VARICOSE VEINS: Swollen and twisted veins visible just under the surface of the skin. They appear most commonly in the legs, but also can develop in other parts of the body.

VARIATION: The quality of being made of many different coloured patches, spots or streaks in plant leaves, stems, petals or other parts as a result of the absence of pigment or different combinations of pigments in the affected area. It is caused by genetic differences between the cells of the affected part.

VARIETAL RESISTANCE (HOST-PLANT RESISTANCE): A particular plant's ability to resist injury caused by herbicides, pathogens, or insects. This may be genetically controlled innate or bred phenotypic or physiological property of the plant. Some plants use certain features such as hair covering on their leaves or plants with a thick leaf cuticle to deter organisms.

VARIETY: 1. A taxonomic subdivision of a species consisting of naturally occurring or selectively bred populations or individuals that differ from the remainder of the species in certain minor characteristics. It consists of a group of individuals that differ distinctly from but can be interbred with other varieties of the same species. Examples of varieties include breeds of domestic animals and the races of humans. Varieties of a species generally have distinguishing characteristics such as a flower colour and may arise naturally or through deliberate plant breeding. 2. In mathematics, the set of solutions of a system of polynomial equations, over the real or complex numbers.

VARIMAX ROTATION: A method used in factor analysis for which the aim is to rotate the vector of factors so as to find a few major key rotations that will simplify the analysis.

VARIOMETER: 1. A variable inductor used to measure inductance as part of an alternating current bridge. It consists of two coils connected in series which can move relative to each other. 2. Any of several instruments for detecting and measuring changes in the geomagnetic fields, especially variations in the Earth's magnetic fields. 3. Also known as **rate-of-climb indicator**; a **vertical speed indicator (VSI)** or a **vertical velocity indicator (VVI)**, it is any flight instrument in an aircraft used to inform the pilot of the near instantaneous (rather than averaged) rate of descent or climb.

VARVE: A layer or series of layers of sediment deposited annually in a still body of water such as saline or fresh-water lakes, for example, by a glacier. Varves occur in repetitive series and thus comprise vertical sequences of annual cyclic deposits and can be counted back to date a specific layer. Varves range in thickness from less than a millimetre to over a metre. **Varve dating** or **geochronology** is an absolute technique of dating, using thin sedimentary layers of clays.

The varves have alternative light and dark bands or a pair of thin sedimentary beds; one coarse and one fine representing winter and summer depositions. By counting varves, it is possible to establish an absolute time scale for fossils as old as 20,000 years.

VAS DEFERENS (SPERM DUCT): A partially coiled tube or duct in the male reproductive system that transports matured sperm from the epididymis (a tube attached to the testis where sperm is stored) to the urethra that carries semen and urine outside the body. There are two ducts, connecting the left and right epididymis. Each vas ends by joining the duct of a seminal vesicle, to enter the urethra (the urinary passage), where that traverses the prostate gland. It is a thick-walled tube, about 0.61 metres long that begins at the tail of the epididymis and eventually joins the duct of the seminal vesicle to form the ejaculatory duct. When each of the two vas deferentia is sealed to prevent sperm from mixing with the semen produced by the male during ejaculation to prevent conception, it is called **vasectomy**.

VAS EFFERENS: Any of various small duct carrying sperm from the testis to the epididymis.

VASA RECTA (STRAIGHT ARTERIES OF KIDNEY; VASA RECTA RENIS; STRAIGHT ARTERIOLES OF KIDNEY): Thin-walled blood vessels that branch from the efferent arterioles leaving each glomerulus in the vertebrate kidney. They form U-shaped loops adjacent to the loop of Henle and eventually drain into the blood renal vein. On the descending portion of the vasa recta, sodium chloride (NaCl) and urea ($\text{CO}(\text{NH}_2)_2$) are reabsorbed into the blood, while water is secreted. On the ascending portion of the vasa recta, sodium chloride and urea are secreted into the interstitium, while water is reabsorbed.

VASCULAR BUNDLE (FACICLE): A unit of xylem and phloem tissues or a strand of combined phloem and xylem tissues within young plants making up the transport system. The xylem and phloem are separated by cambium in plants that undergo secondary thickening.

VASCULAR CAMBIUM: A layer of lateral meristematic tissue between the xylem and phloem in the stems of woody plants. It is the source of both the secondary xylem (inwards, towards the pith) and the secondary phloem (outwards) and is located between these tissues in the stem and root. It consists of cells that are partly specialised, for the tissues that transport water solutions, but have not reached any of the final forms that occur in their branch of specialisation. When these cells have divided and specialised further they make up the secondary vascular tissues, secondary xylem and the secondary phloem.

VASCULAR CYLINDER (STELE): A central column formed by the vascular tissue of a plant root. It is surrounded by parenchymal ground tissue.

VASCULAR PARENCHYMA: Specialised parenchyma cells in the phloem of plants. It is the parenchyma, which is found associated with the vascular tissues xylem and phloem. Accordingly, it is distinguished into xylem parenchyma and phloem parenchyma. Parenchyma is mainly involved in functions like storage and respiration. It also takes part in other functions like photosynthesis, absorption, secretion and protection.

VASCULAR PLANTS: Plants such as ferns, gymnosperms and flowering plants containing vascular tissue, which is a conducting system consisting mostly of phloem (food-conducting tissue) and xylem (water-conducting tissue).

VASCULAR SYSTEM: 1. In plants, the tissues that carry and circulate sap through the plant. Xylem tissue transports water and dissolve minerals to the leaves, and phloem tissue conducts food from the leaves to all parts of the plant. 2. In animals, the network of vessels and tissues that carry and circulate fluids such as blood and lymph through the body. The blood vascular system consisting of the heart and blood vessels circulate respiratory gases, nutrients, and other metabolites into and out of the cells.

VASCULARISATION: The formation of blood vessels into a tissue with the result that the oxygen and nutrient supply is improved. Vascularisation of tumours is often a prelude to more rapid growth. It is thought to be triggered by angiogenesis factors that stimulate endothelial proliferation and migration.

VASECTOMY: A simple surgical operation to make a male sterile. The vas deferens from each testis is cut and tied to prevent passage of sperm from the testes to the prostate gland, to cause sterility or prevent infection during ejaculation.

VASELINE (PETROLATUM): A high-boiling hydrocarbon residue from the distillation of petroleum. It is a jelly used for lubrication and as an emollient.

VASOACTIVE INTESTINAL PEPTIDE, VIP (POLYPEPTIDE): A peptide hormone containing 28 acid residues discharged by endocrine cells of the upper part of the small intestine in response to the entry of partially digested food from the stomach. VIP along with secretin stimulate the pancreas to produce a thin watery secretion containing bicarbonate. This raises the pH in the intestine in preparation for secretion of pancreatic enzymes. It is produced in many areas of the human body including the gut, pancreas and suprachiasmatic nuclei of the hypothalamus in the brain having an effect on several different parts of the body. It also inhibits gastric secretion. It occurs as neuropeptide in central nervous tissue, where it functions as a cotransmitter and has a range of other physiological roles. In humans, the vasoactive intestinal peptide is encoded by the *VIP* gene.

VASOCONGESTION: The swelling of bodily tissues caused by increased vascular blood flow and a localised increase in blood pressure.

VASOCONSTRICTION: The narrowing or reduction in the internal diameter of small blood vessels, usually arterioles. Factors causing vasoconstriction include pain, loud noises, fear, a drop in temperature and a fall in blood pressure. In the case of arterioles, the constriction is caused by the action of nerves on the smooth muscle fibres of the arteriole walls.

VASODILATION: The increase in the internal diameter of blood vessels, especially arterioles or capillaries resulting in an increase in blood flow to the area supplied by the vessel. It is caused by the relaxation of involuntary muscles, causing a decrease in blood pressure. Factors causing vasodilation of blood vessels supplying muscles include exercise, high external temperatures, stress and a rise in blood pressure.

VASODILATOR: Any substance that causes widening of blood vessels, permitting free flow of blood. Vasodilator drugs are used to treat high blood pressure (hypertension), angina (chest pain caused by inadequate blood supply to heart muscles), peripheral vascular disease (poor blood flow in the limbs) and heart failure. They include ACE inhibitors, calcium channel blockers, nitrate drugs and sympatholytic drugs.

VASOMOTOR CENTRE: The region of the medulla oblongata of the brain that contains neurones, which mediate the diameter of blood vessels and hence control blood pressure.

VASOMOTOR NERVES: The nerves of the voluntary nervous system that regulate the contraction of smooth muscles in the walls of blood vessels, thereby regulating blood vessel diameter and blood flow.

VEAL: The meat from a young calf. It is usually pale greyish white in colour, firm and fine-grained, with a velvety texture. Calves bred to yield veal are usually raised indoors under controlled temperatures and intensively fed on milk, high-protein calf meal or both.

VECTOR: 1. An organism, such as a mosquito or tick that passively transmits disease causing microorganisms from one animal or plant to another or from an animal to man. 2. A bacteriophage, plasmid or other agent that is used in genetic modification to transfer a segment of foreign DNA into a bacterium or other cell. The foreign DNA is spliced into the vector's DNA, which contains the genes necessary for switching on replication and transcription of the foreign DNA in its new

setting. 3. Also called **Euclidean vector** or **vector quantity**, it is a quantity that has both direction and magnitude. Force and velocity are examples of vectors. Euclidean vector is frequently represented by a line segment with a definite direction or graphically as an arrow, connecting an initial point *A* with a terminal point *B*. It is denoted by $\overrightarrow{AB}$, indicating a movement from point *A* to point *B*. Vectors can also be denoted by boldface letters, for example, **w**, denotes a vector. The magnitude of a vector is a scalar, denoted by normal letters, for example, *w*. Vertical bars surrounding a boldface letter such as **|w|** denote the magnitude of a vector. A vector can be added together (vector addition), subtracted (vector subtraction) and multiplied by scalars (scalar multiplication). The product of a vector and a scalar is a vector. **Resultant vector** is the addition of two or more single vectors together. **Equilibrant** is the vector which has the same magnitude but opposite direction to the resultant vector. Two vectors are equal if they have the same magnitude and the same direction. 4. The course taken by an aircraft or a missile. 5. A computer array of any length, but with only one dimension.

VECTOR ANALYSIS: The application of the techniques of analysis and algebra to the study of vectors. Since vector quantities have both magnitude and direction, they may be represented geometrically by lines with specified directions, whose lengths represent the magnitudes of the vectors concerned; the magnitude of a vector **V** is written **|V|**. The dot (or scalar) product of two vectors **A** and **B** inclined at angle *c* to each other is defined as **A.B = |A||B| cos c**. The cross (or vector) product of two vectors **A** and **B** inclined at an angle *c* to each other is defined as **A × B = (AB sin c)n** where **n** is a unit vector whose direction is perpendicular to the plane of **A** and **B** such that, looking along the direction of **n**, a clockwise turn of less than 180° is required to bring **A** into **B**.

VECTOR ANGLE: In Cartesian geometry, the angle between the position vector of a point and the *x*- or polar axis.

VECTOR FIELD: A set of vectors such that every point in a particular region of space *R* is associated with a single vector. A familiar example of a vector field in nature is a magnetic field.

VECTOR FILE: A collection of objects, shapes and colours, along with their starting and ending points that enables the images to be recreated.

VECTOR FUNCTION: A function whose domain is a subset of *n*-dimensional Euclidean space.

VECTOR GRAPHICS: A method of creating graphic images on a computer, comprising of paths, which are defined by a start and end point, along with other points, curves and angles along the way. A path can be a line, a square, a triangle or a curvy shape. These paths can be used to create simple drawings or complex diagrams. Paths are even used to define the characters of specific typefaces. Unlike Joint Photographic Experts Group (JPEGs), Graphics Interchange Formats (GIFs) and bit-mapped graphics (BMP) images, vector graphics are not made up of a grid of pixels.

VECTOR NORM: For an *n*-dimensional vector *x*, it is a non-negative function which satisfies the following:

- (i) $|x| > 0$, when $|x| \neq 0$ and $|x| = 0$, when $x = 0$;
- (ii) $|kx| = k \times |x|$ for any scalar *k*;
- (iii) $|x - y| < |x| + |y|$.

VECTOR PROCESSING: In computer science, it is a type of parallel processing, which speeds up the processing of information by a computer with pipelined units performing arithmetic operations on uniform, linear arrays of data values and a single instruction involves the execution of the same operation on every element of the array. A **vector processor**, also known as **array processor** is a central processing unit (CPU) that implements an instruction set containing instructions that operate on one-dimensional arrays of data called vectors.

VECTOR PRODUCT (CROSS PRODUCT): The product of two vectors **A** and **B**, with components a_1, a_2, a_3 and b_1, b_2, b_3 , respectively. This can be expressed as

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}. \text{ It can also be written}$$

as $\mathbf{A} \times \mathbf{B} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}$. The vector product is itself a vector, perpendicular to both $\mathbf{A}$ and $\mathbf{B}$ and of length $AB\sin\theta$, where θ is the angle between $\mathbf{A}$ and $\mathbf{B}$. The cross product of two vectors is a vector rather than a number, unlike the dot product. The determinant notation is a useful scheme for remembering the

$$\text{cross product. If } \mathbf{A} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix},$$

$$\text{then the determinant notation is } \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}. \text{ This symbol is actually}$$

not a determinant, due to the fact that $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are vectors while the other elements are numbers.

VECTOR QUANTITY (EUCLIDEAN VECTOR): Any physical quantity with both magnitude and direction. Examples are velocity, acceleration and momentum.

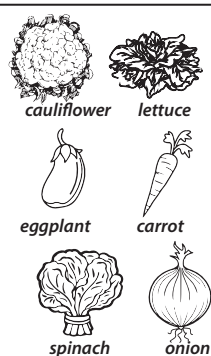
VECTOR SPACE: A set of generalised vectors and a field of scalars, for which an operation of addition is defined so that if $\mathbf{v}_1$ and $\mathbf{v}_2$ are vectors, the sum of $\mathbf{v}_1 + \mathbf{v}_2$ is also a vector.

VECTOR SUM: 1. The binary vector operation that yields a vector of which the length and direction are represented by the diagonal of a parallelogram of which the sides represent the given vectors. This operation is associative and commutative. 2. The result of applying this operation to a sequence of vectors.

VECTOR TRIPLE PRODUCT: In mathematics, the triple product is a product of three vectors. Triple product is used for two different products, the scalar-valued scalar triple product and, less often, the vector-valued vector triple product. The product of three vectors in three space is defined by $\mathbf{A}(\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$ and $\mathbf{A}(\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$.

VECTOR-VALUED: Of a mapping, taking values in an appropriate vector space as contrasted with a scalar-valued mapping taking values in the corresponding scalar field.

VEGETABLE: An edible plant that is grown for its leaves, roots, fruits, stems, seeds and flowers. Vegetables are eaten fresh or cooked as part of main meals or as snacks. Examples include the root of the beet, the leaf of spinach or the flower buds of broccoli or cauliflower, cabbages, beans, onions and carrots. Some common vegetables are illustrated here:



Examples of vegetable

VEGETABLE CROP (VEGETABLE):

A classification of agricultural crops under horticulture. They are plants having edible parts that are used in either cooked or raw state. They are a source of many nutrients, including potassium, fibre, folate or folic acid and vitamin A, E and C, with additional benefits being provided by such vegetable crops as tomatoes, spinach, broccoli and garlic.

VEGETAL: Processes in living things that are "plant-like", especially processes that do not occur through sexual reproduction.

VEGETAL POLE: The part of an egg, where the cells grow and divide much more slowly than do the cells in the animal pole. It is opposite to the animal pole and usually marks the centre of the protoplasm.

It contains much yolk and little cytoplasm and divides much slowly and into larger blastomeres than that of the animal pole and gives rise to the hypoblast of the embryo.

VEGETARIAN: A person who eats no meat for humane, religious or for health sake. Vegetarians get their protein from a variety of sources, such as foods from the large family of legumes, for example, peas or beans. Vegetarians fall into several categories: (i) **Vegans**, who do not consume products of animal origin at all; (ii) **Fruitarianism** eat only fruit and not other vegetables. (iii) **Lacto-vegetarians** consume milk and milk products; (iv) **Ovo-lacto-vegetarians** eat eggs as well as vegetables. (v) **Pescetarians** eat fish but not meat.

VEGETATION: 1. Plants generally and collectively or the mass of plants growing in a particular place. 2. An outgrowth from a body part such as on the membranes surrounding the heart.

VEGETATIVE: 1. Plant parts such as stems, leaves, roots and buds but not flowers, fruits and seeds. 2. Relating to, involving or typical of the growth and maintenance of an organism, rather than its sexual reproduction, especially in plants, in which new individuals develop asexually from specialised structures such as bulbs, rhizomes or runners rather than from specialised sex cells. 3. Characterised by the reduction or absence of the usual mental or physical functions, often as a result of injury to the brain. Such condition is called vegetative state, in which the cerebral cortex, which is the thinking, feeling part of the brain is damaged and does not work properly. The brain stem is not affected and ensures that there are long periods of eye opening (unlike coma) and that there is spontaneous breathing (unlike brain death). Such a person lives a life devoid of mental effort or intellectual interests. Vegetative state may result from an acute brain damage from severe head injury or failure of oxygen supply to the brain.

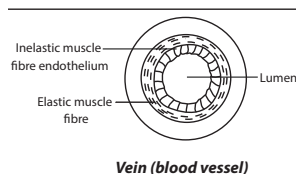
VEGETATIVE NERVOUS SYSTEM (AUTONOMIC NERVOUS SYSTEM, ANS; VISCERAL NERVOUS SYSTEM): The part of the body's nervous system that regulates involuntary activities such as heart rate, digestion, respiration rate, salivation, perspiration, diameter of the pupils, urination and sexual arousal. The autonomic nervous system comprises two antagonistic sets of nerves, the sympathetic and parasympathetic nervous systems. The **sympathetic nervous system** stimulates the heart, dilates the bronchi, contracts the arteries and inhibits the digestive system, preparing the organism for physical action. The **parasympathetic** has the opposite effects and prepares the organism for feeding, digestion and rest.

VEGETATIVE NUCLEUS (VEGETATIVE CELL): The large nucleus formed within the pollen grain of angiosperms along with one or two smaller generative nuclei. After germination of the pollen grain it migrates to the tip of the pollen tube where it may be termed the tube nucleus. It is thought to control the growth and development of the pollen tube and disintegrates when the pollen tube penetrates the nucellus.

VEGETATIVE PROPAGATION (VEGETATIVE REPRODUCTION; VEGETATIVE MULTIPLICATION; VEGETATIVE CLONING): A form of asexual reproduction in plants in which new individuals arise from plant parts such as roots, stems and leaves without production of seeds or spores from specialised multicellular structures such as tubers, suckers, bulbs, rhizomes, etc., which eventually detaches from the parent plant. Artificial methods of vegetative propagation include budding, grafting and making cuttings. Vegetative reproduction also describes asexual reproduction in animals such as budding in Hydra.

VEHICLE: 1. An inactive substance with which a drug is blended to make it easier to administer, apply or take. 2. A powered structure, device or rocket used to transport a payload or another craft through space. 3. A wheeled conveyance used on land for carrying goods or people. 4. A substance such as linseed oil or an acrylic vinyl polymer in which a pigment is mixed for painting.

VEIN: 1. Any blood vessel that transports deoxygenated blood from the body to the heart. They have thinner walls than arteries and contain valves which keep blood flowing in one direction. Most veins carry deoxygenated blood from the tissues back to the heart. Exceptions are the pulmonary and umbilical veins, both of which carry oxygenated blood to the heart. 2. A distinct strand of tissue in a leaf that contains the phloem and xylem. It comprises one of several bundles of vessels and associated tissues of vascular bundles and forms part of a network, arranged in a characteristic pattern. 3. A hard tube that strengthens the wing of an insect or a hollow supporting structure in the wing of an insect that carries blood vessels, nerves and air tubes supplying the wing. 4. Also known as **hydrothermal vein**, it is a deposit of crystallised mixture of minerals such as metallic ores and gemstone in a crack in rock precipitated through the action of hydrothermal activity.



VELAMEN: A whitish spongy multiple epidermis of dead empty cells surrounding the aerial roots of epiphytic plants such as certain orchids. It absorbs moisture from the atmosphere.

VELOCITY: The vector quantity that designates the speed and direction in which a body moves or the time rate of displacement of a body or a measurement that describes both the speed of an object and its direction of travel. Its unit is metres per second (m/s or ms⁻¹) and its symbol is $\vec{v}$ or $\vec{v}$. It is expressed as: $\vec{v} = \frac{\vec{s}}{t}$, where $\vec{s}$ is the displacement and t is the time interval.

VELOCITY GRADIENT: The difference in velocity between adjacent layers of a fluid is moving through a pipe or a solid object. It is given by $\Delta v / \Delta x$, where Δv is the velocity difference between the layers and Δx is the distance between the layers. It is given by $\Delta u / \Delta y$.

VELOCITY POTENTIAL: A scalar function, f , such that the velocity is $\nabla \Phi$ in an irrotational motion.

VELOCITY RATIO (VR): 1. The ratio of the distance moved by the point of application of an effort in a simple machine such as a lever to the distance moved by the point of application of the load at the same time or the ratio of the distance moved by the effort to the distance moved by the load. Velocity Ratio = Distance moved by effort / Distance moved by load. 2. The ratio of the speed of tidal current at a subordinate station to the speed of the corresponding current at the reference station.

VELUM: 1. The membranous flap of tissue partially covering and protecting the sporangium of *Isoetes*. 2. Any of various veil-like bodily structures, such as the soft palate. 3. Any of various membranous structures, such as the ciliated oral membrane of certain mollusc larvae or the veil-like membrane running around the rim of a jellyfish.

VELUTINOUS: Covered with dense, short, soft, spreading hairs; velvety.

VELVETLEAF (*Abutilon theophrasti*): A summer annual broadleaf plant in the family, Malvaceae that grows to about 1.5 metres tall. Native to southern Asia, it was introduced to North America in the mid- 18th century as a potential fibre crop, but has become an invasive species in agricultural regions of the eastern and midwestern United States. The leaves are cordate, covered with short, soft and simple hairs and alternate. The leaf has a unique odour when crushed. The flowers are yellow or orange with five petals that are fused at the base, stalked and found singly or in clusters where the leaf stalk meets the stem. The fruit is a pod-like capsule consisting of a cup-like ring formed by 12 to 15 woody segments. At maturity each segment releases 1-3 seeds through a vertical slit on the outer face of the capsule. The seeds are purplish brown, brown or black ranging from almost triangular to

kidney-shaped. The plant harbours pest and reduces light penetration to crop plants and therefore, considered as stubborn weed for crops like corn, cotton and soybeans in the United States.

VERMERONASAL ORGAN, VNO (JACOBSON'S ORGAN): An olfactory sense (organ of smell) found in many land dwelling vertebrates but not apparent in humans. They are typically paired blind-ending sacs opening into the oral or nasal cavities. They respond especially to pheromones, chemical messengers that carry information between individuals of the same species which influence their behaviour.

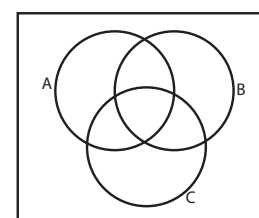
VERMIFUGES (ANTHELMINTICS): A group of anti-parasitic drugs or herbs used to expel or kill parasitic worms such as flukes, tapeworms, nematodes, flatworms, and round worms, and other internal parasites from the body. Examples of such herbs include pawpaw (seeds, roots, leaves, etc.), aloe vera, neem, garlic, parsley and clove. Anthelmintic herbs are safe to use and have few or no side effects.

VENA CAVA: Either of the two large veins: the superior (anterior) vena cava or precaval vein; and the inferior (posterior) vena cava or postcaval vein. They deliver oxygen-depleted blood to the right side of the heart. The superior vena cava returns blood returning from the upper body and head and the inferior vena cava returns blood from below the chest to the heart.

VENATION: 1. Also known as **veination**, it is the arrangement pattern of veins (vascular bundles) in a leaf. Monocotyledons have parallel venation while dicotyledons have reticulate venation. 2. The arrangement of the veins in an insect's wing, which is relevant in its classification.

VENEER: A very thin sheet of wood, such as mahogany, ebony and oak cut in decorative patterns and applied to the surface of cheaper materials to give them an improved appearance.

VENN DIAGRAM (SET DIAGRAMS): A diagram representing a mathematical set or sets and the logical relationships between them. The sets are drawn as circles with an area of overlap between two circles (sets) containing elements that are common to both circles and thus representing a third set. Circles that do not overlap represent sets with no elements in common. It is named after John Venn (1834-1923), English logician.



A Venn diagram representing the sets, A, B, and C

VENOMOUS LIZARDS: Lizards occurring in southwestern United States and Mexico belonging to the family Helodermatidae, which possess venom glands and inject venom when they bite. The only known examples are the gila monster (*Helderma suspectum*) and the beaded lizard (*Heloderma horridum*).

VENT: 1. An opening, as in a wall, serving as an outlet for air, smoke, fumes or the like. 2. The excretory opening of the digestive tract in animals such as birds, reptiles, amphibians and fish through which all eggs and waste material pass. 3. An opening in the Earth's crust from which volcanic material or gases escape. 4. An opening on the ocean floor that emits hot water and dissolved minerals. 5. The opening usually at the centre in a parachute canopy which stabilises it by allowing the air to escape at a controlled rate. 6. Opening to the atmosphere from, for example, a fuel tank.

VENTER: 1. The swollen base of the female reproductive organ of ferns, mosses, liverworts and archegonium in which the egg

cells (oosphere) develop. 2. A part in lower forms of animal life corresponding to the abdomen of mammals. 3. The prominent fleshy portion of a muscle.

VENTILATION: 1. The process by which continuous exchange of gases is maintained across the respiratory surfaces. The ventilation or respiration rate of an animal is the volume of air breathed per minute, which is given by the tidal volume times the number of breaths taken per minute. A respirometer is used in measuring the rate of respiration. 2. The supply of air motion in a space by circulation or by moving air through the space. Ventilation may be produced by any combination of natural or mechanical supply and exhaust. Such systems may include partial treatment such as heating, humidity control, filtering or purification and, in some cases, evaporative cooling. More complete treatment of the air is generally called **air conditioning**.

VENTILATION CENTRE: A set of neurons in the medulla oblongata of the brain that regulates ventilation. The partial pressure of carbon dioxide in the blood and the pH of the blood are monitored by chemoreceptors in the arteries. The ventilation centre responds to a rise in the amount of carbon dioxide in the body by increasing the rate of breathing.

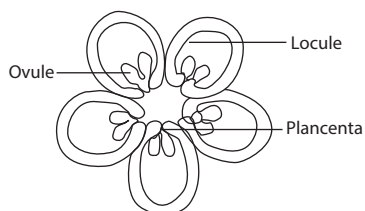
VENTRAL: 1. Located on or affecting the lower surface of an animal's body or the front of the human body. 2. Relating to or situated on or close to the abdomen. 3. Of or on the lower or inner surface of an organ that faces the axis.

VENTRAL AORTA: The artery trunk in the vertebrate embryos that carries blood between the heart and the first aortic arch in embryos. In adult fish it branches into different branchial arteries supplying the gills, while in adult tetrapods, it is represented by the ascending part of the aorta.

VENTRAL CANAL CELL: One of the products that is formed along with the egg cell when the primary ventral cell of an archegonium divides. It has no cell wall and lies at the base of the neck. When the archegonium is mature the ventral canal cell becomes mucilaginous and may produce chemicals to attract the male gametes.

VENTRAL FIN: 1. A fin such as a pelvic fin or an anal fin, located ventrally below the pectoral fins of a fish. 2. A vertical, stabilising fin on the lower rear portion of a fuselage that helps increase directional stability, especially at higher Mach numbers, by increasing the area behind the vertical axis.

VENTRAL PLACENTATION (MARGINAL PLACENTATION): Arrangement of ovules along the ventral suture of the carpel. There is only one elongated placenta on one side of the ovary and ovules are attached to the fusion line of the carpels' margin.



Marginal placentation of apocarpous gynoecium

VENTRAL ROOT: The spinal nerve root that arises from the anterior section of the spinal cord emerging from the lower surface of the spinal cord in animals and from the front surface in humans and mammals. It consists of motor fibres. The ventral root and a dorsal root unite to form a spinal nerve. The ventral root (motor root) of each spinal nerve consists of axons from motor neurons whose cell bodies are found within the grey matter of the spinal cord.

VENTRAL SCALE: Scales found on the underside of a snake's body.

VENTRAL SUTURE: The line of fusion where the margins of the megasporophyll join to form the characteristic tubular shape of the ovary as found in angiosperms. This is often one of the first lines of weakness along which dehiscence occurs at fruit ripening.

VENTRICLE: 1. A space within the brain in which cerebrospinal fluid is produced. 2. One of two lower chambers of the heart that receives blood from the left atria and send it to the body (left ventricle) or the lungs (right ventricle). The right ventricle has thick muscular walls and pumps blood to the lungs while the left ventricle has thicker muscular walls and pumps blood to the rest of the body. Mammals, birds and reptiles have two ventricles which pump deoxygenated blood to the lungs and oxygenated blood to the rest of the body respectively while fishes and amphibians have a single ventricle. 3. Any of the four fluid-filled and linked chambers in the brain of vertebrates with two being in the cerebral hemispheres, one in the medulla oblongata and the other in the posterior part of the fore-brain. The fluids in the ventricles are cerebrospinal fluid filtered from the blood by the choroid plexus.

VENTRICOSE: Inflated or swollen on one side only, as in some corollas, especially in the genus *Penstemon*.



Ventricose corolla

VENTURI TUBE: A device consisting of two tapered sections of pipe joined by a narrow throat for measuring the rate of fluid flow through a pipe or tube. Fluid passing through the tube speeds up as it enters the tube's narrow throat and the pressure drops.

VENULE: 1. A small blood vessel consisting of an endothelial tube enclosed in a variable amount of elastic and collagenous tissue that receives blood from the capillaries and transports it to a vein. 2. A small branching vein in an insect's wing or in a leaf.

VENUS: The second planet in distance from the Sun, with a mean distance of 108.21×10^6 km and mean diameter of 12,103.6 km. It is also the fourth smallest planet in the solar system. It is seen from Earth as a bright morning or evening star. Its atmospheric mass is almost a hundred times that of the Earth. Its atmosphere is mostly carbon dioxide and a small percentage of nitrogen; with an extensive cloud layer of concentrated sulfuric acid. Its surface temperature is 457°C (730 K). It has its orbit between Mercury and the Earth and rotates with a period of 243 days from east to west, in the opposite sense of most other planets.

VERDIGRIS: 1. A green patina of basic copper salts formed on copper, brass and bronze exposed to air or seawater for long periods of time. Its composition varies depending on the atmospheric conditions, but includes the basic carbonate $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$, the basic sulfate $\text{CuSO}_4 \cdot \text{Cu}(\text{OH})_2 \cdot \text{H}_2\text{O}$ and sometimes the basic chloride $\text{CuCl}_2 \cdot \text{Cu}(\text{OH})_2$. 2. A blue or green powder consisting of basic cupric acetate used as a paint pigment and fungicide.

VERIFICATION: 1. The process of checking data being input into a computer to make sure it has been accurately copied from a source document. 2. The establishment of the truth or correctness of something by investigation or the evidence that proves something is true or correct. In science, it involves the use of empirical data, observation, test, or experiment to confirm the truth or rational justification of a hypothesis. Scientific beliefs must be evaluated and supported by empirical data.

VERMICULITE: A natural clay mineral with the chemical formula $(\text{MgFe,Al})_3(\text{Al,Si})_4\text{O}_{10}(\text{OH})_2 \cdot 4\text{H}_2\text{O}$ that expands from 18 to 25 times its original volume with application of heat to 300°C (573 K). It is formed by the weathering of the mica mineral biotite. It is used in many areas including fireproofing, soil conditioning, insulation, etc.

VERMIFORM APPENDIX (APPENDIX): A narrow blind-ended tube leading from the large intestine's caecum near its junction with the small intestine. It contains cells of the immune system.

VERMIN: Various small animals or insects, such as rats, foxes or cockroaches, harmful to plants, birds and other animals or parasitic insects such as lice sometimes found on the bodies of human beings and other animals.

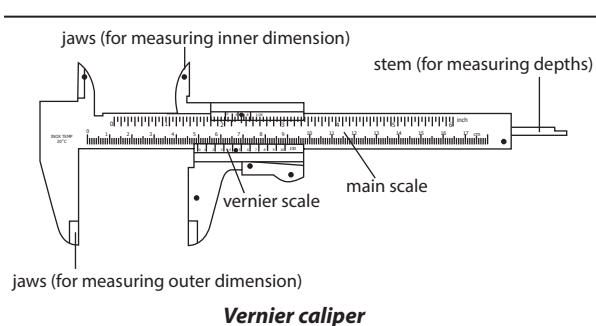
VERNAL: Flowering or appearing in the spring.

VERNAL EQUINOX (SPRING EQUINOX): 1. The time when the Sun having a northerly motion crosses the celestial equator and day and night are of equal length. In the Northern Hemisphere this occurs around March 21 and in the Southern Hemisphere around September 23. It marks the beginning of spring in the temperate zones in the Northern Hemisphere. 2. The point on the celestial sphere where the path of the Sun crosses the celestial equator, in the constellation Pisces.

VERNALISATION (VERNALIZATION): The application of a cold treatment to a germinating seed or seedlings to ensure flowering or the stimulation of flowering by exposure to low temperature. It concerns the first of the three phases of flower formation in plants. In the second stage, for which a certain photoperiod frequently is required, flowers are initiated. In the third stage flowers are unfolded. For example, winter cereals will not flower unless subjected to a period of chilling early in their development, thus they are sown in autumn for flowering the following year.

VERNATION (PTYXIS): The pattern of folding and rolling of leaves in a bud. Many types of vernation are recognised, which may be useful diagnostically. The leaf margins may be rolled forwards to the upper side of the leaf (involute) or rolled backwards to the adaxial surface (revolute). The margin of one side may be wrapped around the other (convolute) or each leaf may be folded in a u-shape, enclosing the next youngest leaf (conduplicate). There may be many longitudinal folds in the lamina (plicate), as in certain palms or the lamina may be rolled from apex to base (circinate), as seen in most ferns.

VERNIER CALIPER (VERNIER): A measuring instrument consisting of an L-shaped frame with a small movable graduated scale parallel to a larger graduated scale, calibrated to obtain more precise readings from the larger scale. It measures internal and external dimensions and also depth by a probe that is attached to the movable head and slides along the centre of the body. It gives direct reading of the distance measured to high accuracy; and can make measurements as small as 0.1 mm or 0.01 cm. It was invented by Pierre Vernier in about 1630.



Vernier caliper

VERRUCOSE (TUBERCULATE): Having a surface covered with small warty projections as, for example, the rind of ridge cucumbers. Pollen covered in wartlike structures (verrucae), such as that of ivy (*Hedera helix*), is normally described as verrucate.

VERSATILE: 1. Referring to a joint or body part such as an insect's antenna, the toe of an owl or the loosely attached anther of a flower that can turn or move freely in more than one direction. 2. Equipment that is robust and can be used for many operations or measurements.

VERSED SIGN: In mathematics, it is the trigonometric function that equals to 1 minus the cosine function.

VERSION CONTROL SYSTEM (VCS): An application, desktop or web-based system that lets developers keep their source code and binary and resource files stored away along with information about different versions. It is used to manage multiple versions of computer files and programs. A version control system or VCS, provides two

primary data management capabilities. It allows users to lock files so they can only be edited by one person at a time and also track changes to files.

VESICLE: 1. A small sac or cyst, or a small bladderlike cavity, especially one filled with fluid. 2. A circumscribed elevation of the epidermis containing serous fluid; a blister. 3. A seminal vesicle is a pouch or sac in many male invertebrates and lower vertebrates that is used for storing sperm. 4. Either of a pair of glands that secrete the fluid component of semen into the ejaculatory duct in males. 5. A spherical structure formed intra-cellularly, by certain arbuscular mycorrhizal fungi. 6. A small, usually spherical cavity in a rock or mineral, formed by expansion of a gas or vapour before the enclosing body solidified.

VESTIGIAL: The part of an animal that does not develop during evolution, and is underdeveloped and is no longer functional or serves no important function. Examples in humans are tailbone (coccyx), wisdom teeth, and appendix.

VERTEBRA: The basic unit of the vertebral column, which is a bone of the spinal column, typically consisting of a thick body or a bony arch enclosing a hole for the spinal cord and stubby projections that connect with adjacent bones or the separate bones in vertebrates that make up the backbone. There are 33 vertebrae in the human vertebral column, which surround and protect the spinal cord and provide some type of axial support for the body.

VERTEBRAL ARTERY: Major arteries of the neck which branch from the subclavian arteries and unite to form the basilar artery in a complex called the **vertebrobasilar system**, which supplies blood to the posterior part of the circle of Willis and thus significant portions of the brain.

VERTEBRAL COLUMN (BACKBONE; SPINAL COLUMN; SPINE): A flexible column of bones (vertebrae) and/or cartilages separated by intervertebral disks and held together by muscles and tendons that extend from the cranium to the coccyx or the end of the tail, encasing the spinal cord and forming the supporting axis of the body. It is the main longitudinal supporting structure and encloses and protects the spinal cord. It also provides the main skeletal support as well as attachment for the muscles of the back. It articulates with the skull by means of the atlas.

VERTEBRATE: An animal with a backbone or vertebral column or an animal with a segmented spinal column, a well-developed brain and a distinct well-differentiated head. It belongs to the subphylum Vertebrata (Craniata), a primary division of the phylum Chordata that includes the fishes, amphibians, reptiles, birds and mammals.

VERTEX: 1. A point on a polyhedron shared by three or more sides or the point farthest from a figure's base or a point of intersection of two sides of a plane figure or the two rays of an angle. 2. The point of intersection of a pencil of lines. 3. For an ellipse, either of the points at which the major axis intersects the curve. 4. One of the nodes that, together with associated edges that makes up a graph. 5. In a parabola, it is the measurement of the maximum or minimum point in a curve. 6. The highest point of a body part, especially the skull or crown of the head. 7. A point on the celestial sphere toward which or from which a group of stars appears to move.

VERTEX FORM: In mathematics, it is an expression for a conic, obtained by a suitable change of variables, in which the vertex is taken as the origin of the coordinate system and the axis of the conic lies along the x -axis.

VERTICAL: 1. A description of a line or plane which is at a right angle with the Earth's surface or to another line or plane. 2. In botany, it refers to being positioned lengthwise, in the same direction as the

axis; leaves positioned with the blade perpendicular, so that neither surface is obviously the upper or lower. 3. The squeezing of a graph or figure towards the x -axis.

VERTICAL ANGLES: The pair of equal angles between two intersecting straight lines. They are opposite angles.

VERTICAL ASYMPTOTE: A vertical line corresponding to a value for which a function is undefined.

VERTICAL IONISATION: A process whereby an electron is removed from a molecule in its ground or an excited state so rapidly that a positive ion is produced without change in the positions or momenta of the atoms. The result is often in an excited state.

VERTICAL LINE TEST: The vertical line test can be used to determine if a relation is a function. If a vertical line can be drawn that crosses two points on the graph of the relation, then the relation is not a function.

VERTICAL POLARISATION: One type of linear polarisation in which the electric waves vibrate or oscillate from top to bottom as the wave travels and its electric field vector of electromagnetic wave is perpendicular to the earth; caused by having antennas vertical to the earth. The position of electric and magnetic field vectors are very important in radio wave communication. This is very important when choosing and installing an antenna as the direction of electric field vector specifies polarisation of the antenna.

VERTICAL RAY: Energy from the Sun that strikes the Earth at a right angle. Vertical rays produce more light and heat than slanting rays.

VERTICAL SECTION OF SOIL (SOIL PROFILE): The soil profile showing the nature and sequence of the various layers from top (Horizon O, organic material at the surface) through to the bottom (Horizon R, the underlying bedrock, such as limestone, sandstone or granite).

VERTICALLY OPPOSITE ANGLES: Angles formed when two straight lines intersect at a point; the resultant four angles formed add up to 360° . In the diagram shown, a and d as well as b and c are vertically opposite angles.

VERTICAL STRETCHING/COMPRESSION: **Vertical stretching** is the stretching of the graph away from the x -axis. **Vertical compression** is the squeezing of the graph towards the x -axis. Compression is a stretch by a common factor less than 1. For the parent function $y = f(x)$, the vertical stretching or compression of the function is $af(x)$.

VERTICIL: A circular arrangement of similar parts such as flowers, leaves or hairs around a central point of attachment; a whorl.

VERTICILLASTER (FALSE WHORL): The arrangement, resembling a complete whorl of flowers that results when two dichasia are opposite each other at a node on an extended rachis. It is seen in members of the Labiatae, for example the white deadnettle (*Lamium album*).

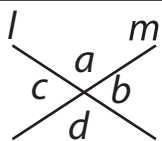


Verticillaster of a plant

VERY HIGH FREQUENCY

(VHF): A radio frequency in the range of 3×10^8 to 0.3×10^9 Hz, which is equivalent to a wavelength in the range 1 to 10m.

VERY HIGH LEVEL LANGUAGE (VHLL): Problem oriented programming languages that require little social training of the user. They are easier to read, write and maintain than machine languages.



a and d are vertically opposite angles, also b and c.

Vertically opposite angles

VERY LARGE ARRAY (VLA): The largest and most advanced radio telescope array in the world, standing at an elevation of 2124 m above sea level. It consists of 27 parabolic dishes, each 25 m in diameter, along three equally spaced arms forming a Y-shaped array, with each arm approximately 21 km long. It is located on the Plains of San Augustine, 80 km west of Socorro, New Mexico.

VERY LARGE-SCALE INTEGRATION, VLSI: The current technology that enables hundreds of thousands of transistors of electronic components to be etched onto a single integrated circuit that is just a few square millimetres.

VERY LOW FREQUENCY (VLF): A radio frequency in the range of 30 kHz to 3 kHz, which is equivalent to a wavelength of 10 to 100 km.

VERY SMALL APERTURE TERMINAL (VSAT): Small earthbound antennas or satellite dishes that enable data to be transmitted between multiple locations over a broad area.

VESICLE: 1. A small usually fluid-filled membrane-bound sac within the cytoplasm of living cell. It occurs for example, as part of the Golgi body, as lysosomes or as micro bodies. 2. A serum-filled blister formed in or beneath the skin. 3. A small cavity formed in volcanic rock by entrapment of a gas bubble during solidification.

VESICULAR-ARBUSCULAR MYCORRHIZA: A form of endotrophic mycorrhiza in which the fungus lives between the cells of the cortex and forms temporary hyphal projections that penetrate the cortical cells. The projections may simply be swollen vesicles or may consist of finely branched masses called arbuscules. Many grasses and crop plants form such associations and the fungi involved are often species of Pythium or Endogone. When soil nutrients are in short supply plants forming such mycorrhizae fare better than uninfected plants.

VESPERTINE: With flowers opening in the evening or night; or functioning in the evening.

VESSEL: 1. Any of various tubular structures such as a blood vessel or lymphatic vessel through which substances are transported. 2. A cell type found in most flowering plants and some ferns, which transports water and dissolved minerals from the roots to the shoots and leaves of the plant, forming part of the sap-conducting tissue xylem. 3. The hollow receptacle, such as a bottle, bucket and bowl, used as a container, especially for liquids. 4. A ship or large boat.

VESSEL ELEMENT: One of the cell types found in xylem, which is an elongated wide dead lignified cell with the end walls either widely perforated or totally broken down and joined end-to-end with similar cells to form a tract for rapid transport of water and inorganic ions in angiosperms. They are absent in most gymnosperms such as the conifers.

VESTIBULAR APPARATUS: The anatomical structures concerned with the vestibular portion of the eighth cranial nerve, which includes the saccule, utricle, semicircular canals, vestibular nerve and vestibular nuclei of the ear. The apparatus is continuous with the cochlea. It consists of three semicircular canals, which detect movement of the head and the utricle and saccule, which detect the position of the head.

VESTIBULAR CANAL: The division of the spiral canal of the cochlea lying above the spiral lamina and the vestibular membrane in the inner ear that connects with the oval window. It contains perilymph, through which pressure waves are transmitted from the oval window via Reissner's membrane to the endolymph within the cochlea duct.

VESTIBULAR SENSE (EQUILIBRIUM SENSE): The mammalian ability to remain upright in relation to the gravity and to detect changes in position and momentum. The principal organs of equilibrium are contained in the inner ear consisting of the utricle, which transmits orientation information and the semicircular canals, which are concerned with acceleration and deceleration. These systems help humans locate their bodies in space.

VESTIBULAR WINDOW (FENESTRA OVALIS; OVAL WINDOW):

A membrane-covered opening between the middle ear and the vestibule of the inner ear to which the base of the stapes is connected and through which the ossicles of the ear transmit sound vibrations to the cochlea. Vibrations coming into contact with the tympanum are transmitted through the middle ear by the three ear ossicles and transmitted to the inner ear by the oval window which is connected to the third ear ossicle. Vibrations from the tympanum reaching the oval window are amplified over twenty times.

VESTIBULE: 1. A cavity or space in the body that serves as the entrance to another canal or cavity, for example, the part of the mouth between the teeth and lips or, the lead from the vulva into the vagina in the mammalian reproductive system. 2. The middle cavity of the inner ear between the cochlea and the semicircular canals.

VESTIBULOCOCHLEAR NERVE (ACOUSTIC NERVE; AUDITORY NERVE):

The eighth cranial nerve that transmits information from the inner ear to the hindbrain, which it enters between the pons and medulla. It serves two distinct functions: The acoustic portion carries sensations of sound and pitch from the organ of hearing, the cochlea; these are relayed to the auditory part of the cerebral cortex; The vestibular portion carries postural sensation from the semicircular canals of the inner ear to the cerebellum. It provides information about position.

VESTIGIAL: Becoming functionless or degenerate in the course of time. A **vestigial organ/structure** is a rudimentary structure or part of an organ such as the appendix in humans that has diminished in size during evolution because the function it served decreased in importance or became totally unnecessary; or an organ that remains in diminished form after it has ceased to have any significant function in the adult organism. Examples of such structures in humans are body hair, tailbone (coccyx), wisdom teeth, and body hair.

VESTITURE: The collective epidermal coverings of a plant.

VETERINARIAN: A person trained and qualified in the medical treatment of animals. A veterinarian diagnoses, treats or researches medical conditions and diseases of pets, livestock and animals in zoos and laboratories. Anything relating to the diseases and treatment of animals is described as **veterinary**.

VEXILLUM: In botany, it refers to the upper and usually largest petal of a papilionaceous flower as in peas and sweet peas.

VGA (VIDEO GRAPHICS ARRAY): A graphics display monitor or interface used in most PCs, developed in 1987 by IBM. In text mode, VGA monitors have resolutions of 720 x 400 pixels for texts, 640 x 480 with 16 colours or 320 x 200 with 256 colours for graphic displays. The total palette of colours is 262,144. Super VGA (SVGA) monitors have resolutions of either 800 x 600 or 1,024 x 768 pixels.

VIABILITY TEST: A test to determine the proportion of living individuals, cells or organisms in a sample. They are often performed on cultured cells and frequently depend on the ability of living cells to exclude a dye (an exclusion test) or to specifically take it up (inclusion test).

VIABLE: Able to exist or capable of developing and surviving without outside help. Examples are a plant that is able to germinate or develop normally or a foetus that can survive outside the womb.

VIABLE BUT NONCULTURABLE: Referring to microorganisms that cannot be cultured on artificial media.

VIABLE COUNT (TOTAL VIABLE COUNT, TVC): A measure of the number of living cells within a culture or the count actually representing the number of colony forming units (cfu) per gram or per ml of the sample.

VIAGRA: The trademark name for the drug sildenafil citrate that is approved by the US Food and Drug Administration (FDA) in March

1998 for treating erectile dysfunction (impotence) in men. When a man gets sexually excited, the nervous system in the erectile tissue of the penis releases nitric oxide (NO), which stimulates an enzyme, phosphodiesterase to produce increased cyclic guanosine monophosphate (cGMP). This relaxes the smooth muscle cells resulting in increased dilation of blood vessels to increase blood flow to the penis to achieve an erection. Its side effects include headache, fainting, indigestion, nasal congestion and impaired vision, etc. It may cause inhibition of phosphodiesterase in the eye, resulting in damage to the retina.

VIBER: A voice over IP (VoIP) and instant messaging app for mobile devices that allows for the exchange of audio, video and images between users. It was developed and released by Viber Media on December 2, 2010.

VIBRATING-STRING EQUATION:

The equation $\frac{\partial^2 y}{\partial t^2} = \frac{T}{\rho} \frac{\partial^2 y}{\partial x^2}$, where x is the direction in which a vibrating string is stretched, y is the displacement, t is time, T is the tension in the string and ρ is its density.

VIBRATION: 1. Continuing periodic oscillation with respect to a fixed reference point or a single complete oscillation. Examples are a ruler 'twanged' over the edge of a table and the motion of a pendulum bob. Vibration is a mechanical phenomenon, whereas oscillations occur about an equilibrium point. The oscillations may be periodic such as the motion of a pendulum or random such as the movement of a tire on a gravel road. Vibration may be desirable such as the motion of a tuning fork, the reed in a woodwind instrument or harmonica or the cone of a loudspeaker, necessary for the correct functioning of the various devices. It can also be undesirable, wasting energy and creating unwanted sound or noise. For example, the vibrational motions of engines, electric motors or any mechanical device in operation are typically unwanted. Such vibrations can be caused by imbalances in the rotating parts, uneven friction, the meshing of gear teeth, etc. Sound and vibration are closely related. Sound or pressure waves are generated by vibrating structures, for example, vocal cords. These pressure waves can also induce the vibration of structures, for example, ear drum. 2. Oscillation of atoms about their equilibrium positions within a molecular system.

VIBRATIONAL REDISTRIBUTION: Intramolecular redistribution of energy among the vibrational modes usually giving a statistical distribution of their populations. It is characterised by the 'vibrational temperature'. For large molecules, this process does not require collisions.

VIBRATIONAL RELAXATION: A process in which a polyatomic molecule in an excited vibrational state returns to a lower vibrational state in the same electronic state by colliding with other molecules.

VIBRATIONAL SPECTROSCOPY: The spectroscopic investigation of the vibrational energy levels of molecules. In the infrared region of the electromagnetic spectrum vibrational transitions are accompanied by rotational transitions.

VIBRATIONAL TERM: Vibrational energy divided by the product of the Planck constant and the speed of light.

VIBRIO: Any of a group of aquatic, comma-shaped bacteria in the family Vibrionaceae, which is the causal agent of cholera and acute bacterial diarrhoea. They are generally Gram-negative, motile and aerobic and are widely distributed in soil and water.

VIBRIO CHOLERA (COMMA BACILLUS; KOCH'S BACILLUS):

A gram negative comma-shaped bacterium with a polar flagellum that causes cholera in humans, the life threatening aspects of which are caused by the exotoxin. They adhere to intestinal epithelium and produce enzymes such as neuraminidase and proteases that facilitate access of the bacterium to the epithelial surface.

VIBRISSAE (WHISKERS): Stiff hairs or modified feathers found in the face of some mammals, which stimulate nerve centres in the skin when they vibrate and used as organs of touch.

VIBRONIC COUPLING: Interaction between electronic and vibrational motions in a molecular entity.

VIBRONIC TRANSITION: A transition which involves a change in both the electronic and vibrational quantum numbers of a molecular entity, as opposed to purely electronic or purely vibrational transitions. The transition occurs between two states, just as in a purely electronic transition, but involves a change in both electronic and vibrational energy.

VICARIANCE: The splitting up of an original biota into several isolated biotas (vicariants) by past geological or climatic events. The vicariants then develop independently and different species evolve. The process is presented as an alternative to the traditional theories of speciation by dispersal that have previously been used to explain biogeographical patterns.

VICENARY: In mathematics, having or using base 20.

VICINAL (VIC): Describing a molecule in which two atoms or groups are linked to adjacent atoms. For example, 1,2-dichloroethane ($\text{CH}_2\text{ClCH}_2\text{Cl}$) is a vicinal or vic dihalide and can be named vic-dichloroethane.

VICINITY: In mathematics, it is a member of a uniformity.

VICIOUS CIRCLE: 1. In logic, a futile form of reasoning in which a conclusion is inferred from premises of which the truth cannot be established independently of that conclusion. 2. In logic, it is explanation given in terms that cannot be understood independently of what was to be explained. 3. A situation arising from self-reference in which some statement is shown to entail its negation and vice versa, such as "this statement is false", which is true only if false and false only if true.

VICTOR MEYER'S METHOD: A method used in measuring vapour density, devised by Victor Meyer (1848-1897), German organic chemist. A weighed sample in a small tube is dropped into a heated bulb with a long neck. Vapourisation takes place and air is displaced, which is collected over water and the volume measured. The vapour density can then be calculated.

VIDEO CAMERA: A portable television camera that takes continuous moving pictures electromagnetically on magnetic tape and generates a signal for display or recording. It captures images by breaking them down into a series of lines. It produces an electrical output signal corresponding to rapid line-by-line scanning of the field of view. The output is recorded on video cassette and is played back on a television screen via a videotape recorder.

VIDEO CARD (GRAPHICS ACCELERATOR; DISPLAY ADAPTER; GRAPHICS CARD; VIDEO ADAPTER; VIDEO BOARD): An internal circuit board that allows a display device such as a monitor to display images from the computer. Video cards can speed up both 2D and 3D graphics rendering. Programs such as photo editors and Web browsers may benefit from 2D acceleration, while CAD design programs and video games will most likely benefit from the card's 3D acceleration. Some programs rely so heavily on the video card that they will not run if a supported video card is not installed.

VIDEO ELECTRONICS STANDARD ASSOCIATION (VESA): An industry standards organisation that set video display and bus standards. VESA was originally founded by NEC and is most well-known for the VL Bus or VESA local Bus standard.

VIDEO GRAPHICS ARRAY (VGA): A graphics display monitor or interface used in most PCs, developed in 1987 by IBM. In text mode, VGA monitors have resolutions of 720×400 pixels for texts, 640×480 with 16 colours or 320×200 with 256 colours for graphic displays. The total palette of colours is 262,144. Super VGA (SVGA) monitors have resolutions of either 800×600 or $1,024 \times 768$ pixels.

VIDEO RANDOM ACCESS MEMORY (VRAM): A form of random access memory (RAM) that allows simultaneous access by two different devices, so that graphics can be handled at the same time as data are updated. VRAM improves graphic display performance.

VIDICON: A vacuum tube containing a photosensitive area or target and an electron gun to read the signal from the target. The silicon target consists of a two-dimensional array of Si-photodiodes having a common cathode and isolated anodes. Irradiation of the target causes the production of electron-holepairs which, by recombination, leads to a depletion of the surface charge.

VIETE'S FORMULA (VIETA'S FORMULA): The formula for π , derived from the infinite product for $\frac{\pi}{2}$, thus

$$\sqrt{\frac{1}{2}} \times \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}} \times \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}} \times \dots$$

The formulae relate the coefficients of a polynomial to sums and products of its roots. It is named after François Viète (1540-1603), French mathematician, also known as Franciscus Vieta, who proved the formulae for positive roots only.

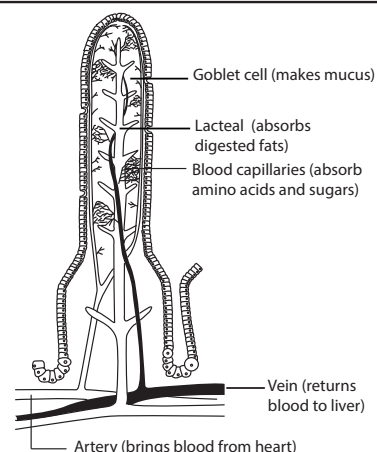
VIGESIMAL: Based on intervals of 20. It is the base-20 notational system for representing real numbers. The digits used to represent numbers using vigesimal notation are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F, G, H, I and J. A base 20 number system was used by the Aztecs and Mayans. The Mayans compiled extensive observations of planetary positions in base-20 notation.

VIGNA UNGUICULATA: The scientific classification for the asparagus bean. They are good sources of protein, vitamin A, vitamin C thiamin, riboflavin, iron, phosphorus, potassium, folate, magnesium and manganese.

VILLIAUMITE: A carmine, isometric mineral formed of sodium fluoride, NaF. It is a rare halide mineral composed of sodium fluoride, NaF. It is very soluble in water and some specimens fluoresce under long and short wave ultraviolet light. It has a Mohs scale hardness of 2.5 and is usually red, pink or orange in colour. It is toxic to humans and must be handled with care.

VILLOUS (VILLOSE): Bearing long, soft, shaggy, hair but unmatted. **Villosullous** is the diminutive of villous.

VILLUS: 1. **Intestinal villus** is microscopic fingerlike outgrowth of the small intestine extending into the lumen of the small intestine and increasing the surface area for absorption of food. 2. **Chorionic villus** is one of the fingerlike projections of the chorion, which develops from the mammalian placenta into the uterine wall and thus increasing the area of contact between maternal and embryonic tissues. 3. A fine, hair-like epidermal outgrowth.



Longitudinal section of intestinal villus

VINCA (PERIWINKLE): A genus of subshrub or herbaceous plants of the dogbane (Apocynaceae) family with a slender creeping stem that includes the periwinkle (*Vinca minor*), also known as lesser periwinkle and *Vinca major*, commonly known as big periwinkle, big leaf periwinkle, large periwinkle or greater periwinkle. Periwinkle is native to Europe and western Asia and is usually found in North-west Africa, South-west Asia and Europe. It grows from 20-70 centimetres

tall. The stem frequently develops roots when it comes into contact with the ground, enabling the plant to spread widely for the leaves to receive maximum light for photosynthesis. It bears leaves that have opposite leaf arrangement and are simple broad lanceolate to ovate. The flowers are violet with gamopetalous petals. It is used in cemeteries as an evergreen maintenance-free ground cover. It is also used in treating childhood leukaemia and lymphomas as it contains vincristine that is administered as a chemotherapy agent.

VINCULUM: 1. The horizontal bar (—) placed over two or more terms in an algebraic expression to indicate that they are to be treated as a single term, for example, $x + y - z$. It is also used to denote a radical ($\sqrt[n]{ab + 2}$), repeating decimals (0.111), the line segment $\overline{AB}$ joining two points A and B , complex conjugate $\bar{z} = \overline{x + iy} = x - iy$ or negation of a logical expression, $\overline{A \wedge B} = \overline{A} \vee \overline{B}$. 2. A ligament that limits the movement of an organ or part.

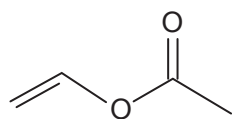
VINE: 1. A plant that supports itself by twining, climbing or creeping along a surface or a flexible, weak-stemmed plant whose long, slender, fast-growing shoots rely on other plants or objects for support. In addition to winding or twining around supporting structures, vines may use thorns, tendrils or hooks to climb and anchor to a structure and maximise exposure to sunlight without expending energy on supporting tissue. There are four general types of vines: (i) **Lianas**, which are woody perennials with rope like stems that climb high into tree crowns where they produce flowers and fruit. They are leafless along the stem and may grow to more than 150 metres; (ii) **Scramblers** spread quickly over neighbouring vegetation with the aid of spines or thorns, ensuring that they receive maximum available sunlight and moisture. Examples are blackberry, rose, ivy and bougainvillea. (iii) **Trailing vines** are weak-stemmed herbs, which spread over the ground until they find a suitable supporting structure such as a fence, pole or tree. Examples are varieties of bean plants. (iv) **Creeping vines** lie on the ground, often sending roots into the soil from nodes on the stems and spreading slowly across the ground. 2. A video sharing social media application that allows users to share 6-second video clips with their followers. It was founded in June 2012 and later acquired by Twitter in October 2012, just before its official launch.

VINEGAR: An acidic liquid produced from the fermentation of ethanol in a process that yields its key ingredient, acetic acid or ethanoic acid. It is made by the action of bacteria on wine or cider or a dilute acetic acid made by fermenting beer or wine. Natural vinegars also contain small amounts of tartaric acid, citric acid and other acids. It is used widely in the food industry for preserving food.

VINEMEOUS: With long, flexible twigs; composed of twigs or having a twig-like appearance.

VINYL ACETATE (ETHENYL ACETATE; ETHENYL ETHANOATE):

A colourless, water-insoluble, flammable liquid with the chemical formula $C_4H_6O_2$, density 0.934 g/cm^3 , melting point -93°C (180 K) and boiling point 72.7°C (346 K). It is a monomer from which the plastic and adhesive polyvinyl acetate (PVA) are made. It is prepared industrially by the reaction of ethylene and acetic acid with oxygen in the presence of a palladium catalyst. It is also prepared by the gas-phase addition of acetic acid to acetylene.



Chemical Structure
of Vinyl Acetate

VINYL GROUP: The double-bonded organic group $\text{CH}_2=\text{CH}-$, which are derivatives of ethene, $\text{CH}_2=\text{CH}_2$, with one hydrogen atom replaced with some other group. It is used as a chemical intermediate and in the production of polymers and copolymers, for example, the polyvinyl resins.

VINYLATION: The process of introducing the vinyl into a compound by reaction with acetylene, for example, vinylation of alcohols gives vinyl ethers, such as vinyl ethyl ether.

VIOLACEOUS: Violet-coloured; of, relating to or denoting plants of the violet family (Violaceae).

VIOLAXANTHIN: A xanthophyll pigment that absorbs light in the blue region of the spectrum. Violaxanthin may in certain circumstances serve as a precursor of abscisic acid. It may also, with riboflavin, be involved in the photooxidation of endogenous auxins.

VIOLET: 1. Any of the approximately 500 species of herbaceous plants or low shrubs that make up the genus *Viola*, family Violaceae which includes the small, solid-coloured violets and the larger-flowered, often multicoloured violas and pansies. Examples are begonia, the gourd, the passion flower and the papaya. 2. Bluish-purple colour similar to the colour of wild violets, used as non-selective weed killers.

VIOLOGEN DYES (BIPYRIDILIUM DYES): A group of organic non-selective weedicide that kills herbaceous plants by damaging chlorophyll and interfering with photosynthesis. Because of its high toxicity its use is strictly controlled.

VIPER: Venomous snake belonging to the family Viperidae, with a compact sturdy body, large triangular head and short tail. A typical example is the gaboon viper (*Bitis gabonica*), the snake with the longest fangs in the world.

VIRAL: 1. In biology, relating to, or caused by a virus. 2. In information technology, spreading of information, image, video, etc., especially on the Internet or in emails.

VIRESCENCE: The abnormal condition of becoming green, especially in plant parts that are not normally green.

VIRGATE: Shaped like a wand or rod; straight, slender and erect.

VIRGIN FOREST (OLD-GROWTH FOREST): A matured forest in its natural state, without being exploited or influenced by any human activity. These type of forests are often biologically diverse, and home to many rare species, threatened species, and endangered species of plants and animals.

VIRGIN LANDS PROGRAM: A programme launched in 1954 by the Soviet (USSR) leader Nikita Khrushchev, to increase grain production by ploughing and sowing grain on idle and virgin land that had never been farmed before.

VIRGIN NEUTRON: Any neutron from any source before collision.

VIRIAL EQUATION: A gas law that tries to explain the behaviour of real gases, as opposed to an ideal gas. It has the form $pV = RT + Bp + Cp^2 + Dp^3 + \dots$, where B , C and D are known as virial coefficients. They are characteristic of the interaction potential between the particles and in general depend on the temperature.

VIRION: A term for a virus particle, which consists of a core of DNA or RNA surrounded by a protein coat, sometimes covered by an outer envelope. The core gives the virus infectivity and the capsid provides specificity.

VIROID: Any of various small naked single-stranded RNA molecules which are not enclosed in a protein coat of any kind, resembling a virus but smaller than any of the known viruses and which infect plant cells and cause disease. Viroids appear to be transmitted from one cell to another through cellular debris. Examples of viroids are disease agents such as potato spindle tuber viroid and citrus oxocortis.

VIROLOGY: The scientific study of viruses and viroids and diseases caused by them, which is considered to be a subfield of microbiology or medicine. The study encompasses their structure, classification and evolution, their ways to infect and exploit host cells for virus reproduction, their interaction with host organism physiology and immunity, the diseases they cause, the techniques to isolate and culture them and their use in research and therapy. Many viruses,

such as animal RNA viruses and viruses that infect bacteria or bacteriophages, have become useful laboratory tools in genetic studies and in work on the cellular metabolic control of gene expression.

VIRTUAL: 1. Being something in effect even if not in reality or not conforming to the generally accepted definition of the term. 2. Created, simulated or carried on by means of a computer or computer network.

VIRTUAL CHANNEL IDENTIFIER (VCI): It is used in asynchronous transfer mode (ATM) cell in conjunction with the virtual path indicator (VPI), to indicate where an ATM cell is to travel over a network.

Virtual path indicator (VPI) is an 8-bit header inside each ATM cell that indicates where the cell should be routed. ATM is a method that many Internet Service Providers (ISPs) use to transfer data to client computers. Because ATM sends packets over fixed channels, the data is easier to track than information sent over the standard TCP/IP protocol.

VIRTUAL COMMUNITY: In computer science, it is a group of people sharing common interests, ideas and feelings over the Internet, using the electronic conferencing system. It also includes exchanging useful information regarding real-life events such as illnesses. It includes forums, online chat rooms, specialised information communities, email groups, etc.

VIRTUAL DEVICE DRIVER (VxD): A software file that acts as a buffer between a hardware device (such as the hard disk drive controller, keyboard and serial and parallel ports) and software programs. This helps ensure that all software programs needing access to a hardware device do not conflict with each other. In Microsoft operating systems, it handles software interrupts for the main hardware devices.

VIRTUAL IMAGE: A situation where an image is seen at a point from which the rays appear to come to the observer but do not actually do so or an image in which the outgoing rays from a point on the object always intersect at a point. No image will be formed on a screen placed at this point or it cannot be projected onto a screen or photographic film. An example is a flat mirror, where one's image is perceived at twice the distance from one to the mirror.

VIRTUAL INCUBATOR: A business that does not necessarily have a physical office space but choose to provide service to its 'fledglings' or inductees over the Internet using specially created software packages.

VIRTUAL LEARNING ENVIRONMENT (VLE): A virtual classroom that allows teachers and students to communicate with each other online. Class information, learning materials and assignments are typically provided via the Web. Students can log in to the class website to view this information and may also download assignments and required reading materials to their computers. Some VLEs allow assignments and tests to be completed online.

VIRTUAL MACHINE: An emulation, usually of a program or another operating system that does not physically exist within another platform.

VIRTUAL MACHINE MANAGER (HYPERVISOR): Software program that allows multiple operating systems, called **guests** to share a single computer system, called **host**. The hypervisor manages the system's processor, memory, and other resources to allocate what each guest or operating system requires.

VIRTUAL MEMORY: A memory management technique for creating the illusion that a computer has more memory than it really has by swapping blocks or pages of data between memory and external storage. Systems that employ virtual memory use hardware memory more efficiently than systems without virtual memory and make the programming of applications easier.

VIRTUAL PARTICLE: A subatomic particle that exists for a limited time and space, introducing uncertainty in its energy and momentum due to the uncertainty principle. Because energy and momentum in quantum mechanics are time and space derivative operators, due to Fourier transforms their spans are inversely proportional to time duration and position spans, respectively.

VIRTUAL POINT OF PRESENCE (vPoP): A telephone link which enables users to connect to a distant point of presence (PoP) for the price of a local call. vPoP is in contrast to a point of presence (PoP) which is operated by the Internet provider themselves.

VIRTUAL PRIVATE NETWORK (VPN): A network that creates a secure network connection over a public network such as the Internet or a private network owned by a service provider. There are a number of systems that enables network creation using the Internet as the medium for transporting data. The most common types of VPNs are remote-access VPNs and site-to-site VPNs.

VIRTUAL REALITY: A type of computer simulation of a real or imaginary system in which the user has the impression of being in a physical environment. These systems are used for training and for entertainment.

VIRTUAL REALITY BROWSER (VR BROWSER): In computer science, it is an application that enables users to walk through a virtual reality scene on their monitors.

VIRTUAL REALITY MODELLING LANGUAGE (VRML): In computer science, it is a method of displaying 3-dimensional images on a web page. VRML environments can contain animation, audio and video files and hyperlinks to simulate realistic views of the inside and outside of a building, bridges, tunnels, etc. These views change in real time in response to the computer user's movements or mouse clicks.

VIRTUAL STATE: The state of the virtual particles that are exchanged between two interacting charged particles. These particles, known as photons are not directly observable. They are constructed to enable phenomenon to be explained in terms of quantum mechanics.

VIRTUAL TRANSITION: A non-observable liquid-crystal transition that occurs below the crystallisation temperature and is determined by extrapolation in binary phase diagrams.

VIRTUAL WORK: The total work done by all the forces acting on a system in static equilibrium is zero for a set of infinitesimally small displacements, subject to physical restraints. If the restraints do no work, for example, by acting at right angles to a possible motion, the system is in equilibrium if and only if the virtual work is zero.

VIRTUALBOX (VB): A virtualisation package that is installed on a host operating system to allow additional guest operating system.

VIRULENCE (PATHOGENICITY): The disease producing ability of a microorganism or the quality of being extremely poisonous, infectious or damaging or the extent to which a disease or toxin possesses this quality.

VIRULENT: Capable of overcoming a host's defence mechanisms and causing a disease sometimes of rapid onset and severe symptoms.

VIRUS: 1. A sub microscopic particle of a nucleic acid surrounded by protein that can only replicate within a host cell. Viruses are between 20 and 100 times smaller than bacteria and hence are too small to be seen by light microscopy. Viruses can be found in virtually all life forms, including humans, animals, plants, fungi and bacteria. They cause diseases such as common cold, influenza, polio and small pox. It consists of a core of nucleic acid, either DNA or RNA but never both, enclosed in a protein shell. They are products of organisms but are not organisms and have the following characteristics: they are incapable of neither metabolism nor reproduction; they have no nucleus and cytoplasm. Because viral processes so closely resemble normal cellular processes, abundant information about cell biology and genetics has come from studying viruses. Viral Diseases of animals include Hepatitis C, Indolent, Ebola, Hepatocellular Carcinoma, Corona virus and Lichen planus. Viral diseases of plants include Tobacco Mosaic virus, Tomato spotted wilt, Tomatoes yellow leaf curl, Potatoes virus, cauliflower virus and African Cassava Mosaic. Viruses are sub-microscopic, infectious particles that multiply only inside living host cells. 2. A computer program that is part of another and inserts copies of itself, often damaging the integrity of stored data. It travels with the program that contains it. Examples are the Trojan horse or wormy.

VIRUSOID: A circular single-stranded RNAs dependent on plant viruses for replication and encapsidation that infects plants and consists of a small naked single-stranded circular RNA molecule. Virusoids are similar to viroids in size, structure and means of replication (rolling-circle replication). The virusoid is introduced to its host cell by an infecting plant virus, in which it is carried as a passenger. Virusoids are not considered as viruses but as subviral particles.

VISCERAL: A collective term for the inner part of a structure of an animal or the internal organs that lie in the coelomic cavities of animals.

VISCERAL MUSCLES: Smooth muscles found in the walls of the digestive tract, bladder, arteries and other internal organs. As with any muscle, the smooth, involuntary muscles of the visceral muscle tissue are composed of bundles of specialised cells capable of contraction and relaxation to create movement.

VISCOELASTICITY: The viscous and elastic properties certain materials such as asphalt and many polymers exhibit, when undergoing deformation. Viscous materials, like honey, resist shear flow and strain linearly with time when a stress is applied. Elastic materials strain instantaneously when stretched and just as quickly return to their original state once the stress is removed. There are two types of viscoelasticity: (i) **Linear viscoelasticity**, in which the function is separable in both creep response and load, applicable only for small deformations; (ii) **Nonlinear viscoelasticity** in which the function is not separable. It usually happens when the deformations are large or if the material changes its properties under deformations.

VISCOMETER: An instrument used to measure the viscosity (resistance to internal flow) of a fluid. In the falling-sphere viscometer, based on Stokes' law, the speed of fall of a ball falling through a sample of the fluid can be measured. There are other devices such as the Ostwald viscometer which are used to measure viscosity of fluids. In the Ostwald Viscometer, a bulb in a capillary tube is filled with the liquid and the time taken for the meniscus to reach a mark on the capillary below the bulb is read as a measure of the viscosity.

VISCOSE (CELLULOSE XANTHATE): A sticky yellowish or brown syrupy solution made by treating cellulose with sodium hydroxide and carbon disulfide. The solution is then regenerated as continuous filament for making rayon and cellophane.

VISCOSITY (FLUID FRICTION): A measure of the resistance to flow that a fluid offers, as a result of its internal friction between the molecules. It applies to the motion of an object moving through a fluid as well as the motion of a fluid passing on an object. Viscosity is a major factor in determining the forces that must be overcome when fluids are used in lubrication or transported in pipelines. The viscosity of liquids decreases rapidly with an increase in temperature, while that of gases increases with an increase in temperature. The SI unit for viscosity is the newton-second per square metre (N-s/m²). **Viscous** is a term used to describe a fluid that has a relatively high resistance to flow.

VISCOUS FLUID: A body in which the stress tensor σ is given by $\sigma = -pI + \sigma^E$, where p is the pressure and σ^E is a deviatoric tensor arising from the motion of the body.

VISIBLE: Region of the electromagnetic spectrum extending from about 400 nm to 760 nm. This is the wavelength region to which the human eye is sensitive. There are no precise limits for the spectral range of visible radiation since they depend upon the amount of radiant power reaching the retina and on the responsivity of the observer.

VISIBLE HORIZON (APPARENT HORIZON): A projection of the local horizon outward to intersect with the celestial sphere.

VISIBLE RADIATION: 1. Electromagnetic radiation such as sunlight that falls within the range of wavelengths that can be detected by the human eye. It is from about 400 (violet)-750 (red) Nm. In between violet and red are the following colours: blue (450 Nm, 6.66×10^{14} Hz), cyan (500Nm, 5.99×10^{14} Hz), green (550 Nm, 5.45×10^{14} Hz),

yellow (580 Nm, 5.16×10^{14} Hz) and orange (600 Nm, 5.0×10^{14} Hz). 2. A mechanical device that uses illumination as a signal or warning, especially a beacon or traffic signal.

VISIBLE SPECTRUM: The range of electromagnetic radiations from 400 (violet)-750 (red) Nm to which the human eye is sensitive. Electromagnetic radiation in this range of wavelengths is called **visible light** or simply light. In between violet and red are the following colours: blue (450 Nm, 6.66×10^{14} Hz), cyan (500Nm, 5.99×10^{14} Hz), green (550 Nm, 5.45×10^{14} Hz), yellow (580 Nm, 5.16×10^{14} Hz) and orange (600 Nm, 5.0×10^{14} Hz). The spectrum does not, however, contain all the colours such as pink or magenta that the human eyes and brain can distinguish. Visible light is essential for all living things on Earth. Visible light is essential for all living things on Earth. Without visible light, life would not exist in any form. It represents the peak of the Sun's spectrum and about half of all of its radiant energy. Visible light is essential for photosynthesis, which enables plants to produce the carbohydrates and proteins that are the food sources for animals. Coal and oil are sources of energy accumulated from sunlight in plants and microorganisms millions of years ago and hydroelectric power is extracted from one step of the hydrologic cycle kept in motion by sunlight at the present time.

VISION: The sense of sight, using the sense organ, eye which perceives the form, colour, size, movement and distance of objects or the special sense for light that, gives larger animals an image of the outside world. Visual stimuli are typically rays of light entering the eyes and forming images on the retina at the back of the eyeball. Human vision is most sensitive for light comprising the visible spectrum in the range 400 (violet)-750 (red) Nm or 400-790 THz ($7.5-4.29 \times 10^{14}$ Hz).

VISUAL ACUITY (SHARPNESS OF VISION): A measure of the ability of the eye to distinguish between objects that lie close together. It is the reciprocal of the angular size of critical detail which is just large enough to be seen. Visual acuity depends on the cone cells which are most densely packed in the fovea, close to the centre of the retina and therefore in the optimum position to receive focussed light.

VISUAL BASIC: A third-generation event-driven programming language and development environment created by Microsoft. It is an extension of the BASIC programming language that combines BASIC functions and commands with visual controls. Visual Basic provides a graphical user interface GUI that allows the developer to drag and drop objects into the program as well as manually write program code.

VISUAL CORTEX (STRIATE CORTEX): The region of the cerebral cortex of the brain occupying the entire surface of the occipital lobe, where sensory information from the eyes is interpreted.

VISUAL DISPLAY UNIT (VDU): The part of a computer system or word processor on which text or diagrams are displayed. The monitor comprises the display device, circuitry and an enclosure. The display device in modern monitors is typically a thin film transistor liquid crystal display (TFT-LCD) thin panel, while older monitors use a cathodic ray tube about as deep as the screen size.

VISUAL MAGNITUDE: The apparent magnitude of a celestial body in a wavelength range approximating that of the human eye. This is usually different from photographic magnitude because photographic emulsions do not exactly match the way the human eye's sensitivity varies with wavelength. Visual magnitude is now essentially synonymous with V magnitude, which is determined photometrically.

VITAL CAPACITY: The total amount of air that can be led from the lungs after maximum inspiration. It is equal to the inspiratory reserve volume plus the tidal volume plus the expiratory reserve volume. The maximum is about 4.5 litres in human and in trained male athletes it can be 6 litres.

VITAL STAINING: In microscopy, it is the applications of a non-poisonous dye to a tissue with the aim of improving contrast and making certain parts that cannot be seen under normal staining techniques visible. It is important in viewing cellular organelles. Examples of vital stains include janus green, trypan blue and vital red. Janus green, for example, is used in observing mitochondria.

VITALI COVERING: In mathematics, it is a set E in Euclidean n -space by hypercubes (where the ratio of largest to smallest side remains bounded) with the property that, for every element e of E , there is a member of the cover that contains e and has arbitrarily small positive measure.)

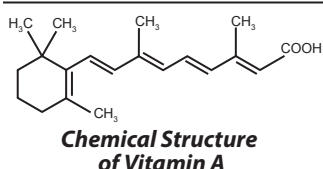
VITALI COVERING THEOREM: The result that if a class of closed hypercubes is a Vitali covering of a set E in n -space, then there is a countable sequence of pairwise-disjoint members of the covering of which the union J has Lebesgue outer measure equal to that of E ; that is, $\mu^*(E \setminus J)$.

VITALI SET: The subset of real numbers that is not Lebesgue measurable. It is constructed by taking one element from each equivalence class of real numbers where the equivalence relation is differing by a rational.

VITALISM (VITALISM THEORY): The belief that natural phenomena are governed by a life force outside the realm of physical and chemical laws. The belief that organic compounds could only be produced in living cells.

VITAMIN: One of the numbers of organic compounds required in comparatively small quantities by living organisms for normal health or for normal functions of the body. There are about 14 generally recognised key vitamins. They are generally classified as water-soluble (Vitamins B and C) or fat soluble (Vitamins A, D, E and K occurring only in animals, especially vertebrates). Only vitamin D can be synthesised by the body. All the others are essential constituents of the diet. Deficiency of one or more vitamins in the nutrient supply results in deficiency diseases. Many vitamins are destroyed by light and heat.

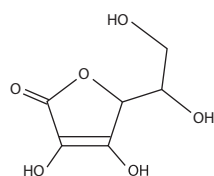
VITAMIN A (RETINOL): A fat-soluble vitamin or a mixture of vitamins, especially vitamin A_1 or a mixture of vitamins A_1 and A_2 , occurring principally in fish-liver oils, milk and some yellow and dark green vegetables and functioning in normal cell growth and development. Green plants contain precursors of the vitamin, notably carotenes that are converted to vitamin A in the intestine wall and liver. Lack of it affects the eye by causing exophthalmia or night blindness, which may eventually result in blindness.



Chemical Structure of Vitamin A

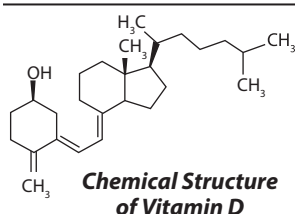
VITAMIN B COMPLEX: A group of 9 water-soluble vitamins that characteristically serves as components of coenzymes. Plants and many microorganisms can manufacture B vitamins but dietary sources are essential for animals. Heat tends to destroy them. The B complex vitamins include B_1 (Thiamin), B_2 (Riboflavin), B_6 (Pyridoxine), B_{12} (Cyanocobalamin or Cobalamin), Nicotonic acid, Pantothenic acid, Biotin and Lipoic acid.

VITAMIN C (ASCORBIC ACID): A colourless crystalline water-soluble vitamin important in animal metabolism. Its key sources are citrus fruit and green vegetables. Most organisms synthesise it from glucose, but humans and other primates must obtain it from their diet. It is essential in collagen synthesis, wound healing, blood-vessel maintenance and immunity. Deficiency of it leads to scurvy. It is readily destroyed by heat and light.



Chemical Structure of Vitamin C (Ascorbic Acid)

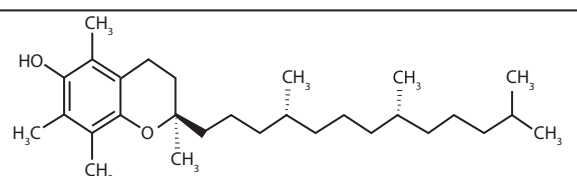
VITAMIN D: A fat-soluble vitamin containing a number of distinct chemicals that enhances the absorption of calcium and phosphorus from the intestine and, with parathyroid hormone, mobilises their deposition in bones. It occurs in the form of two steroid derivatives namely, vitamin D_2 found in yeast and vitamin D_3 occurring in animals.



Chemical Structure of Vitamin D

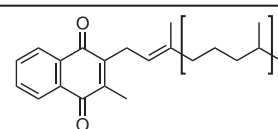
It is stored in the liver and, to a lesser extent, in the fatty tissue in the skin. Vitamin D deficiency causes a loss of muscle tone, restlessness and irritability. It also causes rickets in children and demineralisation and softening of the bones (osteomalacia) in adults. For most adults, the main source of vitamin D is from the action of sunlight on skin.

VITAMIN E (TOCOPHEROL): A fat soluble vitamin made up of several closely related compounds, deficiency of which leads to a range of disorder in different species including muscular dystrophy, liver damage and infertility. The richest sources include wheat germ oil, sunflower oil and roasted peanuts. Vitamin E is an antioxidant and mutually protects vitamins A and C. It prevents the oxidation of unsaturated fatty acid in cell membranes and hence maintaining their structure.



Chemical Structure of Vitamin E

VITAMIN K: A group of compounds derived from 2-methyl-1,4-naphthoquinone that prevents bleeding in mammals and birds. Vitamin K_1 (phyloquinone) is produced by green plants. A related form, vitamin K_2 (menaquinone), is produced by intestinal bacteria. Vitamins K_1 , K_2 and K_3 (menadione) can be chemically synthesised. Deficiency of Vitamins K leads to prolonged bleeding. Reliable sources include egg yolk and green vegetables.



Chemical Structure of Vitamin K

VITICULTURE: The branch of horticulture that is concerned with the study and production of grapes, especially for making wine.

VITREOUS HUMOUR: The transparent colourless fluid that fills the space between the lens and the retina of the eyeball of humans and other vertebrates that is secreted from the ciliary body. It consists of 98% water, a small amount of collagen, salt and sugar. It does not have its own blood supply, as it is not served by any blood vessels. Its functions include transmitting light to the retina, exerting enough pressure to maintain the round shape of the eye so that the lens can focus sharp images on the retina.

VITRIOL (OIL OF VITRIOL): An old name for sulfuric acid. Hydrated iron(II)sulfate was known as green vitriol and hydrated copper(II)sulfate as blue vitriol.

VITTA: A resin canal or oil cavity. Vittae are often found in the fruits of plants in the Umbelliferae. The number and position (whether between or in the primary ridges of the fruit) are important diagnostic characters in the family.

VIVIPARITY (VIVIPAROUS): 1. Also called **placental viviparity**, it is a method of reproduction in animals in which the embryo develops inside the body of the female, from which it gains nourishment. This method of reproduction common in placental mammals such as most mammals, many reptiles, and a few fishes. Viviparous is producing living young which has developed within the body rather than hatching from eggs. 2. A form of asexual reproduction in which the fruit or seed develops into a bud-like structure before detaching from the parent; as occurs, for example, in *Avecinia* (white mangrove), sisal, certain grasses and the spider plant. In many mangroves, for instance, the seedling germinates and grows, while still attached to its parent before dropping into water in order to be transported away.

VIVISECTION: The process of cutting open living animals for experimental purposes, in order to advance physiological and pathological knowledge.

VLF (VERY LOW FREQUENCY): A radio frequency in the range of 30 kHz to 3 kHz, which is equivalent to a wavelength of 10 to 100 km.

VLSI (VERY LARGE-SCALE INTEGRATION): The current technology that enables hundreds of thousands of transistors of electronic components to be etched onto a single integrated circuit that is just a few square millimetres.

VOCAL CORDS: Folds of tissue within a mammal's larynx and a bird's syrinx or a pair of elastic membranes that project into the larynx in air breathing vertebrates. The cords are covered with a mucous membrane and pass horizontally backward from the thyroid cartilage to insert on the smaller, paired arytenoid cartilages at the back of the larynx. Vocal sounds are produced when expelled air passing through the larynx vibrates the cords. Muscles in the larynx change the pitch of the sound by adjusting the tension of the vocal cords, hence the pitch of the sound produced depends on the tension of the cords, which is controlled by muscles and cartilages in the larynx.

VOCAL MATCHING (CALL MATCHING): In zoology, a distinctive vocal interactive process, often displayed by some animal species in which members duplicate each others call. Examples of animal species that display such traits include some birds, for example, parrots, members of the finch family; and some cetaceans. Vocal matching may be done for mating, social or warning purposes.

VOCAL SAC: The flexible membrane of the skin found on each side of the mouth of male frogs and toads that inflates and acts as amplifier for mating calls.

VOICE: Sound produced through the mouth and by the passage of air between the vocal cords. The mechanism for generating the human voice can be subdivided into three parts: the lungs, the vocal folds within the larynx and the articulators. In humans, the sound is much amplified by the hollow sinuses of the face and is modified by the movements of the lips, tongues and cheeks.

VOICE RECOGNITION: Ability of an electronic security device to recognise the unique voice characteristics of a particular person, just as a fingerprint. In contrast, **speech recognition** is the ability to recognise spoken words only and not the individual voice characteristics.

VOID: 1. A space free of matter called vacuum. 2. In heat transfer, void is formed where there is a departure from nucleate boiling, causing a critical heat flux. 3. A space in a soil mass not occupied by solid mineral matter. This space may be occupied by air, water, or other gaseous or liquid material. 4. Void type is a keyword indicating the absence of type information in many programming languages. 5. A large region of space at the centre of a group of galaxy superclusters. 6. Void coefficient is an estimated number in nuclear engineering.

VOID INTERSECTION PROPERTY: The property of a collection of sets that their intersection is empty.

VOID RATIO: The ratio of the volume of void space to the volume of solid particles in a given soil mass, expressed as a decimal. Its symbol is e . Its equation is $e = \frac{V_v}{V_s}$, where V_v is volume of void space

and V_s is volume of solid particles.

VoIP (VOICE OVER INTERNET PROTOCOL; IP TELEPHONE; INTERNET PHONE): An Internet protocol that allows digital telephone connection over the Internet instead of analogue telephone lines. An example of VoIP is Skype. This allows for cheaper and even free long distance or international phone calls.

VOLATILE: 1. A substance such as petrol and ether with low boiling points that readily passes from the liquid to the vapour (gaseous) phase at specified condition of temperature and pressure. On the other hand, a **volatility product** is the product of the concentrations of two or more molecules or ions that react to form a volatile substance. 2. Volatiles are a group of compounds with low boiling points that

are associated with a planet's or Moon's crust and/or atmosphere. 3.

Volatile oil, also known as **essential oil** is oil derived from plants with aromatic compounds. It is used in cosmetics and flavouring.

4. **Volatile anaesthetics** are a class of anaesthetics which easily evaporate. 5. **Volatile acidity** is a term used in winemaking to indicate an unacceptably high level of acid or vinegar and baking soda.

6. **Volatile substance abuse** is the abuse of household inhalants such as some types of glue containing volatile compounds. 7. **Volatile variables** are variables that can be changed by an external process.

8. In computing, **Volatile memory** is the memory that lasts only while the power is on, for example, RAM is volatile, if the computer is turned off anything stored currently in it is erased. Because of RAM's volatile nature, users must frequently save their work to a permanent medium, such as a hard drive, which is non-volatile to avoid losing data if the system's power is interrupted.

VOLATILISATION (VOLATILIZATION): The conversion of a solid or liquid to a gas or vapour by application of heat, by reducing pressure, by chemical reaction or by a combination of these processes. A substance can often be separated from another by volatilisation and can then be recovered by condensation of the vapour. The substance can be made to volatilise more rapidly either by heating to increase its vapour pressure or by removal of the vapour using a stream of inert gas or a vacuum pump. Chemical reactions are sometimes utilised to produce volatile products.

VOLATILISER (VOLATILIZER): Material added to a sample to increase its volatilisation or that of some component of it. Volatilisers increase the fraction volatilised either by forming more volatile compounds or by increasing the total surface area of all analyte particles, for example, by explosive disintegration or by dispersal of the analyte in a highly volatile matrix.

VOLCANIC ROCK (EXTRUSIVE ROCK): Igneous rock that has formed from volcanic activity on the surface of the Earth or under the oceans. Examples of volcanic mountains are Vesuvius, Strombolian and Plinian.

VOLCANO: A naturally occurring opening in the surface of the Earth fissure vent connected to a magma source through which molten, gaseous and solid material is ejected. The molten or liquid rock is known as **lava** when it reaches the surface. Volcanoes occur basically along constructive or destructive plate margins; however, some volcanic activity occurs away from the margins.

VOLCANOLOGY (VULCANOLOGY): The scientific study of volcanism that examines especially the processes, products and hazards associated with active or potentially active volcanoes and how magma and associated gases rise from the Earth's interior and are emitted and the resultant structures. It focuses on eruptive activity that has occurred within the past 10,000 years of the Earth's history, particularly eruptions during recorded history.

VOLT: The SI unit of potential difference and electromotive force defined as the potential difference necessary to produce a current of one ampere through an electric circuit with a resistance of one ohm. It is also equal to one joule of energy per coulomb of charge, J/C. It is named in honour of the Italian physicist Alessandro Volta (1745–1827), who invented the voltaic cell. Its symbol is V.

VOLTA, ALESSANDRO GIUSEPPE ANTONIO ANASTASIO (1745-1827): An Italian physicist, who made voltaic cell, thus providing the first practical source of electric current. He also invented the electrophorus in 1780, a device used to generate static electricity by friction. In 1815, he was appointed director of the philosophical faculty at the University of Padua. The SI unit, voltage was named after him in 1881.

VOLTAGE: A common name for either potential difference or electromotive force expressed in volts, with the symbol V .

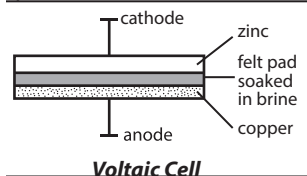
VOLTAGE AMPLIFIER: An electronic device that increases an input signal in the form of a voltage or potential difference delivering an output signal that is larger than the input, by a specified ratio. The voltage gain of the amplifier is the amplitude ratio of the output voltage to the input voltage.

VOLTAGE CLAMP: An experimental apparatus for measuring the flow of ion currents across the membrane channels in cells such as neurons, while holding the membrane voltage at a set level.

VOLTAGE DIVIDER (POTENTIAL DIVIDER): A resistor or a chain of resistors connected in series that can be tapped at one or more points to obtain an output voltage, which is a known fraction of the input voltage across the whole resistor or chain. If R_1 and R_2 are two resistances obtained after tapping and if V is the voltage across the divider and v is required voltage, then $v/V = R_2/(R_1 + R_2)$. It is also a three-terminal resistor with a sliding contact that forms an adjustable voltage divider. If only two terminals are used (one side and the wiper), it acts as a variable resistor or rheostat. It is used to create a reference voltage and may also be used as a signal attenuator at low frequencies.

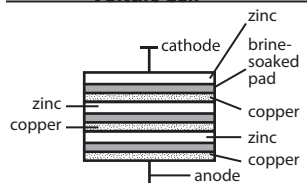
VOLTAGE-GATED ION CHANNEL: Any transmembrane ion channel found along the axon and at the synapse that opens and closes in response to changes in electrical potential across the cell membrane in which the channel is situated. They play a crucial role in excitable neuronal and muscle tissues, allowing a rapid and co-ordinated depolarisation in response to triggering voltage change.

VOLTAIC CELL (GALVANIC CELL): An electrochemical cell that produces electromotive force (emf) by the conversion of chemical energy to electrical energy or as a result of chemical reactions that take place within the cell. These reactions occur at the surfaces of two electrodes which are dipped into an electrolyte. An example is a battery or accumulator.



Voltaic Cell

VOLTAIC PILE: The first electric battery, devised by Alessandro Volta in 1800, consisting of a number of alternating copper or silver and zinc discs (electrodes) separated by cloth or cardboard soaked in brine (electrolyte) to increase the electrolyte conductivity.



Voltaic Pile

VOLTMETER (COULOMETER): A scientific instrument used for measuring the quantity of electricity or electric charge. It is not the same as a voltmeter which measures electric potential. The voltmeter is an electrolytic cell and the measurement is made by weighing the element deposited or released at the cathode in a specified time. The change in mass (m) of the cathode due to the deposition on it of a metal from a solution of its salt, enables the charge (Q) to be estimated from the relationship $Q = m/z$, where z is the electrochemical equivalent of the metal. The SI unit for quantity of electricity is the coulomb, while the SI unit for electric potential is the volt.

VOLTERRA'S INTEGRAL EQUATION: The two integral equations, known as first and second kind linear Volterra equations. The first kind is given by $f(x) = \int K(x,t)y(t)dt$, where f and the kernel, $K(x,t)$ of the integral are given and y is sought. Volterra's integral equation of the second kind is given by $y(x) = f(x) + \lambda \int K(x,t)y(t)dt$.

VOLTMETER: An instrument calibrated in volts that measures the electromotive force or potential difference between two points

in a circuit. Moving coil instruments are widely used. Cathode ray oscilloscopes can also be used as voltmeters. Generally, a galvanometer is used in series with a resistor of high values and to measure an a.c. voltage, a rectifier is included in the circuit. Analogue voltmeters move a pointer across a scale in proportion to the voltage of the circuit; digital voltmeters give a numerical display of voltage by the use of an analogue to digital converter.

VOLUME: The amount of three-dimensional space occupied by a substance such as a solid, liquid, gas or plasma or a body. Solids and fluids have fixed volumes but a gas will have the same volume as the container it occupies. Its symbol is V and its unit is cubic metre (m^3).

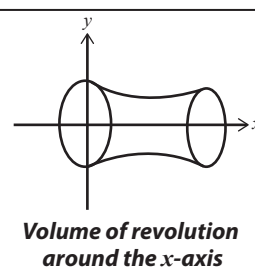
VOLUME FLOW RATE: Volume of a component crossing a surface divided by the time. Its symbol is q_v .

VOLUME FRACTION: Volume of a constituent of a mixture divided by the sum of volumes of all constituents prior to mixing. Its symbol is Φ .

VOLUME INTEGRAL: An integral over a volume, which can involve vectors and scalars. The volume element $dV = dx dy dz$ is a scalar. For a vector function $F(x,y,z)$ the volume integral is given by $\int_V F dV$. For a scalar function $\phi(x,y,z)$ the volume integral is given by $\int_V \phi dV$.

VOLUME OF REVOLUTION: A volume generated by rotating a plane curve about an axis in its plane.

VOLUME OF THE STATIONARY PHASE: The volume of the stationary liquid phase or of the active solid or of the gel in the column. The volume of any solid support is not included.



Volume of revolution around the x-axis

VOLUME STRAIN: The change of volume divided by the original volume. Its symbol is θ .

VOLUME VISCOSITY (DILATATIONAL VISCOSITY): A quantity ζ which enters into equations at any point where the flow involves a change in volume; it is dilatational.

VOLUMETRIC ANALYSIS (TITRIMETRIC ANALYSIS): A procedure used for determining the concentration of a solution, involving quantitative analysis, using accurately measured volumes of solutions to react with a standard solution, by means of titration. A known volume of a solution of unknown concentration is reacted with a solution of known concentration and volume (standard solution) until a reaction end point indicated by a colour change or precipitation is reached. For gases, the main technique is in reacting or absorbing gases in graduated containers over mercury and measuring the changes, while for liquids it involves titrations.

VOLUMETRIC FLASK: A laboratory flask primarily intended for the preparation of definite or fixed volumes of solutions and therefore calibrated for a single volume only. It is made of glass or plastic and consists of a flat bottomed bulb with a long neck, usually fitted with a stopper. The stopper is normally made with a chemically resistant plastic such as polypropylene rather than glass. The neck has a single ring graduation mark and a label.

VOLUMETRIC PIPETTE (PIPETTE): Graduated glass tubing used to measure precise volumes of a solution. The tube is open at the top and bottom and a slight vacuum (suction) at the top pulls liquid into the calibrated section. Breaking the vacuum allows liquid to leave the tube.

VOLUMIC: Attribute to a physical quantity obtained by division by volume. Charge density may thus be called volumic charge which is the electric charge in a section of space divided by the volume of that section.

VOLUNTARY ACTION: An action of the body that occurs under the control of will power.

VOLUNTARY MUSCLE (SKELETAL MUSCLE; STRIPPED MUSCLE; STRIATED MUSCLE): A type of muscle made up of long straight multinucleated fibres attached to the skeleton or bones and bound by sarcolemma and which contracts and relaxes under the control of will. The muscle fibres may be of two main types: **fast twitch fibres**, adapted to produce quick, powerful movements; and **slow twitch fibres**, adapted to slower, endurance type movements. Examples of voluntary muscles are muscles of the chest, neck and abdomen. Involuntary muscles include muscles of the digestive system, smooth muscles, etc.

VOLUNTEER: In botany, it is a plant that has grown from self-sown seed. Volunteers may be important inoculum sources for some diseases.

VOLUTIN: A granular substance that stains easily, found in the cytoplasm of some bacterial and fungal cells including the cyanobacteria. It serves to store phosphates for the energy needs of the cell.

VOLVA: The cuplike structure that encircles the base of the fruiting bodies of many basidiomycete fungi. It is part of the remnants of the universal veil.

VOLVOX: A genus of chlorophytes, a type of freshwater green algae in the form of individual cells with inter-cytoplasmic connections organised into a sphere. It forms spherical colonies of up to 50,000 cells.

VOMERONASAL ORGAN (VNO): An organ of smell found in many terrestrial vertebrates, but not apparent in humans. Vomeronasal organs are typically paired blind-ending sacs opening into the oral or nasal cavities. The entrances to the sacs are from the incisive ducts which communicate with the nasal cavity and a pit in the roof of the mouth, just behind the dental pad. The function of the vomeronasal system is the detection of pheromones, chemical messengers that carry information between individuals of the same species. Vomeronasal system is found in many animals, including all snakes, lizards, mice, rats, elephants, cattle, dogs, cats, goats and pigs.

VON BAEYER NOMENCLATURE: A system of naming polycyclic compounds originally introduced by Adolf von Baeyer in 1900, which has since been extended.

VON LAUE, MAX (1879-1960): German physicist, who won the Nobel Prize in Physics in 1914 for his discovery of the diffraction of x-rays by crystals. In addition to his scientific endeavours with contributions in optics, crystallography, quantum theory, superconductivity and the theory of relativity, he had a number of administrative positions which advanced and guided German scientific research and development during four decades. He was instrumental in re-establishing and organising German science after World War II. He championed Albert Einstein's theory of relativity and studied the quantum theory, the Compton Effect and the disintegration of atoms.

VORTEX DYNAMICS: The branch of fluid mechanics that is concerned with the dynamics of vortices (rotating fluid). There are many applications of vortex dynamics in engineering, atmospheric dynamics and in oceanography. Vortex dynamics is also important in quantum liquids such as superfluid. In the simplest sense, vorticity is the tendency for elements of the fluid to "spin."

VORTEX LINE: A line that is tangent to the vorticity vector at any point and is roughly parallel to the axis.

VORTEX PINNING: The pinning of the magnetic vortices (magnetic flux quanta, Abrikosov vortices) by different kinds of the defects in a type II superconductor. In some practical applications of superconductivity vortex pinning is desirable.

VORTICITY: The curl or rotational of the velocity field describing the continuum motion near some point. It is a vector that measures the local rotation at a point in a fluid, which is equal to twice the angular momentum of a particle of the fluid around which there is circulation.

VORTICITY TENSOR (BODY SPIN; SPIN TENSOR): The skew, symmetric part of the velocity gradient; if $\mathbf{Q}$ is the body spin and $\mathbf{L}$ is the velocity gradient then $\mathbf{\Omega} = 1/2(\mathbf{L} - \mathbf{L}^T)$. This is the local angular velocity expressed in tensor form.

VOUS SYSTEM: The part of the central nervous system that controls the involuntary activities of the smooth muscles of the digestive tract, blood vessels, the heart and the glands. The sympathetic system responds to stress when it speeds the heart rate, increases blood pressure and generally prepares the body for action.

VPN (VIRTUAL PRIVATE NETWORK): A network that creates a secure network connection over a public network such as the Internet or a private network owned by a service provider. There are a number of systems that enables network creation using the Internet as the medium for transporting data. The most common types of VPNs are remote-access VPNs and site-to-site VPNs.

vPoP (VIRTUAL POINT OF PRESENCE): A telephone link which enables users to connect to a distant point of presence (PoP) for the price of a local call. vPoP is in contrast to a point of presence (PoP) which is operated by the Internet provider themselves.

VR BROWSER (VIRTUAL REALITY BROWSER): In computer science, it is an application that enables users to walk through a virtual reality scene on their monitors.

VRAM (VIDEO RANDOM ACCESS MEMORY): A form of random access memory (RAM) that allows simultaneous access by two different devices, so that graphics can be handled at the same time as data are updated. VRAM improves graphic display performance.

VRML (VIRTUAL REALITY MODELLING LANGUAGE): In computer science, it is a method of displaying 3-dimensional images on a web page. VRML environments can contain animation, audio and video files and hyperlinks to simulate realistic views of the inside and outside of a building, bridges, tunnels, etc. These views change in real time in response to the computer user's movements or mouse clicks.

VSEPR THEORY (VALENCE SHELL ELECTRON PAIR REPULSION): A theory for predicting the shapes of molecules based upon the extent of electron-pair electrostatic repulsion. The premise of VSEPR is that the valence electron pairs surrounding an atom mutually repel each other and will therefore adopt an arrangement that minimises this repulsion, thus determining the molecular geometry. The number of electron pairs surrounding an atom, both bonding and nonbonding, is called its **steric number**.

VULCANISATION (VULCANIZATION): A technique for hardening rubber by heating and chemically combining it with sulfur compounds to cause cross-linking of the polymer chains. This process makes the rubber stronger and more elastic.

VULCANITE (EBONITE): A hard black insulating material with a form of rubber as the base. It is made by the vulcanisation of rubber with a high proportion of sulfur. It was formerly used for denture bases.

VULGAR FRACTION (COMMON FRACTION; SIMPLE FRACTION): A fraction having an integer both as a numerator and a denominator. It is a **proper fraction** if its denominator is greater than its numerator, for example, 4/5.

VULNERABILITY: 1. The degree to which a system is susceptible to, or unable to cope with, adverse effects of climate change, including climate variability and extremes. Vulnerability is a function of the character, magnitude, and rate of climate variation to which a system

is exposed, its sensitivity, and its adaptive capacity. 2. In information technology, a flaw in code or design that creates a potential point of security compromise for an endpoint or network.

VULNERABLE SPECIES: A categorisation of threatened species by the International Union for Conservation of Nature (IUCN) and defined as a species that is likely to become endangered unless circumstances such as habitat loss or destruction, threatening its survival and reproduction improve. There are currently 4728 animals and 4914 plants that are vulnerable in the world.

VULNERARAIES: Agents or substances containing properties that heal wounds, burns and fresh cuts. Certain herbs can be effective in doing this; such herbs include marigold, plantain, comfrey and inflammation.

VULVA: The external opening of the vagina or the external female genital organ. It includes labia majora, mons pubis, labia minora, clitoris, bulb of the vestibule, vestibule of the vagina, greater and lesser vestibular glands and the opening of the vagina. The vulva also contains the opening of the female urethra.

VX (O-ETHYL S-[2-(DIISOPROPYLAMINO)ETHYL] METHYLPHOSPHONOTHIOATE): A highly toxic colourless oily liquid with the chemical formula $C_{11}H_{26}NO_2PS$, density 1.0083 g/cm^3 , melting point -50°C (223 K) and boiling point 298°C (571 K). It is a deadly nerve gas several times more toxic than sarin but less volatile. It is an organophosphorous compound and is probably one of the most toxic of all nerve agents. Its production and stockpiling was banned by the Chemical Weapons Convention of 1993.



W BOSON (W PARTICLE): Any pair of elementary particles with either a positive or negative charge (W^+ or W^-) considered to transmit the weak interaction between other elementary particles in similar ways as photons transmit the electromagnetic interaction. It is classified as intermediate vector bosons. W particles have a rest mass of 1.435×10^{-25} kg (806 GeV).

WACKER PROCESS: Commercial organic synthesis for the manufacture of ethanal by the oxidation of ethylene to acetaldehyde by oxygen in the presence of palladium chloride and cupric chloride.

WAD: A small mass of soft material used for keeping things apart or in place, to plug up a hole, padding, stuffing or packing.

WADE'S RULES (WADE/MINGOS RULES; POLYHEDRAL SKELETAL ELECTRON PAIR THEORY): A set of rules for predicting the structure of stable structures of polyhedral inorganic organometallic and organic compounds based on the number of electrons in the framework counted in a particular way. It was originally formulated by K. Wade and further developed and evolved by D. M. P. Mingos and others. The electrons counted are known as **skeletal electrons** and are valence electrons of atoms positioned in the vertices of a molecular polyhedron that form bonding electron pairs of the framework bonds. The Wade/Mingos rules are very accurate in predicting structure when there are 4 electrons for each vertex.

WAGNER-MEERWEIN REARRANGEMENT: A rearrangement in which a hydrogen, alkyl or aryl group migrates from one carbon to a neighbouring carbon. It often occurs to stabilise a carbocation formed as an intermediate during the reaction.

WAHUHA: A pulse sequence in nuclear magnetic resonance (NMR) used to reduce line width. It was named after its inventors Waugh, Huber and Haberland. The sequence effects an averaging procedure by twisting the magnetisation vector in various directions.

WAIS (WIDE AREA INFORMATION SERVER): Software tool formerly used in early internet search engines for indexing large amounts of information and search for and retrieve it from a range of databases on remote computers including the Internet. It has now been replaced by programs such as Google.

WAIST: 1. Part of the body between the rib cage and the hips, usually narrower than the rest of the trunk. The **waistline** refers to the horizontal line where the waist is narrowest or to the general appearance of the waist. 2. The narrow part of an insect's body between the thorax and the abdomen. 3. The middle section of an aircraft's fuselage.

WALD, ABRAHAM (1902-1950): Hungarian mathematical statistician and geometer, who did major work on the problem of the curvature of surfaces, wrote articles on topology and metric spaces, measure and set theory and lattice theory, among others.

WALD'S EQUATION: Let $X_1, X_2, \dots, X_N$ be a sequence of independent identically distributed random variables, each with expected value μ and let N be a random variable, then the expected value of $\sum_{i=1}^N X_i$ is the product of μ and the expected value of N , i.e., $E[X_1 + X_2 + \dots + X_N] = E(X_1)E(N)$.

WALD'S STATISTIC: A statistic of the form: $W = \{g(T)\}D^{-1}g(T)$, where T is an estimator of a vector parameter θ , g is some vector valued function and D is an estimator of the variance-covariance matrix of the vector $\{g(T) - g(\theta)\}$. The statistics are used to test the null hypothesis that $g(\theta) = 0$, where 0 is a vector with all its entries equal to 0. If T is the maximum likelihood estimator, then W has an approximate chi-squared distribution with p (the number of elements in θ) degrees of freedom.

WALDEN'S RULE: An empirical rule suggested by P. Walden (1863-1957) concerning ions in solutions, stating that the product of the molar conductivity Λ_m and the viscosity η , is approximately constant for the same ions in different solvents.

WALK: In mathematics, it is an alternating sequence of edges and vertices in a graph in which the end of each edge (except the last) is the beginning of the next. A **closed walk** is a walk in which the first and last vertices are identical. A **trail** is a walk in which all the edges are distinct and not repeated. Where no vertex is revisited in the course of a trail, it is called a **path**. The special case where the path finishes at the vertex where it started is an exception and is termed a **cycle** or **loop**.

WALL EFFECT: Any effect caused by the ionisation in an ionisation chamber by electrons liberated from the walls of a container on the system it encloses.

WALL PRESSURE: The mechanical pressure exerted by the cell content against the cell wall. The rigidity (turgor) of a plant often depends on the level of wall pressure found in the cells of the stem. A cell absorbing water by osmosis will continue to expand until its selectively permeable membrane and cell wall can stretch no further. The resistance to stretching is what is known as wall pressure.

WALLACE EFFECT: The idea, proposed by the naturalist A. R. Wallace, that those reproductive barriers within a species may be developed and subsequently improved by selection. If there are a number of optimum phenotypes favoured by selection, intermediate organisms less fitted to the environment will be eliminated by natural selection. Thus, hybrids between such phenotypes will be at a disadvantage and hybridisation will be selected against.

WALLACE, ALFRED (1823-1913): British author, collector of wildlife specimens and naturalist who was one of the first to formulate the groundbreaking theory of evolution by natural selection, for which he shares the credit with Charles Darwin. Wallace's studies in South America (1848-1852) and in the Malay Archipelago (1854-1862) of different forms of life led him to his theory of natural selection, one of the most important of all biological theories. Darwin had at the same time independently come to the same conclusion, which they co-jointly read at a meeting of the Linnaean Society of London in 1858. He also studied the geographical distribution of plants and animals in the Malay Archipelago. He traced the boundary, later to be named Wallace's Line, between native regions of Asian and Australian animals which marked the southernmost limit of the Asian mainland before glaciers separated Borneo, Java and other islands from the continent. The mammals in South East Asia are different from and more advanced than their Australian counterparts because the Australian continent had split away from Asia before the better adapted placental mammals evolved in Asia. Examples are the Australian marsupials and monotremes which were able to thrive while those in Asia became extinct due to competition from the better adapted placental mammals.

WALLACE'S LINE: A hypothetical line that runs between the Indonesian islands of Bali and Lombok and represents the separation of the Australian and Oriental faunas.

WALLERIAN DEGENERATION: The characteristic sequence of changes that occur in a vertebrate sensory or motor peripheral neuron after its axon has been severed.